

Geometry, History, and Constraint

A Structural Quantitative Analysis of Physical Form, Flux
Conservation, Sequence Compatibility, and Rank Restoration in
Gravitational Inference

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Scope: Conceptual and methodological study in philosophy of physics, with exploratory public-data analysis and synthetic numerical audits. The numerical cases test identifiability, conditional generation, and falsifiability; they do not establish a galactic spatial metric with ellipsoidal level sets or the absence of dark matter.

Abstract

Physical equations gain predictive economy by compressing source, geometry, inherited state, boundaries, and transition rules. Structural quantitative analysis temporarily separates these explanatory roles and then restores the constraints that make their combinations admissible. The map $R = \Phi_{\text{map}}(\Gamma, S, H, B, I)$ is not an inventory of independent causes: history enters through a sufficient present state or, when required, an explicit memory closure. Sequence compatibility restricts lawful histories but does not itself supply empirical rank.

In the gravitational case, $\varepsilon_{g0,S} \equiv 1/(4\pi G_0)$ is assigned to the reference source-response role, while directional structure is represented empirically by a neutral residual tensor T_R . Promotion to a geometrical T_T requires an independent metric derivation and cross-channel agreement. An exponential-disk control gives $g_{\text{eq}}/g_{\text{pole}} = 1.722$ and a 34.5% spherical-estimator overstatement at $3R_d$, showing that ordinary source geometry must not be counted twice. A flux-conserving finite-volume generator preserves total flux and robustly produces a thinner coupled branch and larger spherical-equivalent apparent mass, but domain and late-time tests restrict the result to finite-time conditional generation.

Exploratory SPARC galaxy-level holdouts find that simple J_R families improve on a baryon-only baseline but do not outperform a fixed radial-acceleration-relation benchmark, which wins 29–30 of 30 repeated comparisons. BTFR-informed calibration is therefore separated from structure-first prediction, and finite flat-looking intervals are not treated automatically as asymptotic. Synthetic conservation–sequence and pseudo-longitudinal audits further show that lawful linkage can reduce parameter freedom while independent observational channels with nonparallel sensitivities are still needed to restore likelihood rank. The framework consequently distinguishes analytical separability, numerical reproducibility, empirical identifiability, predictive success, geometrical explanation, and ontological establishment.

Keywords: structural quantitative analysis; physical form; historical inheritance; sequence compatibility; gravitational response; tensor lattice; flux conservation; generative simulation; identifiability; SPARC; baryonic Tully-Fisher relation

1. Introduction: From Quantitative Laws to the Form of Physical Phenomena

1.1 The explanatory question

Modern physics turns force, mass, energy, field strength, and curvature into compact quantitative relations. That success leaves a second question open: which parts of an observed form arise from the source, spatial or spacetime structure, inherited state, boundaries, and the interaction presently acting? A

formula may compress all of these into one coefficient or operator. Computational compression is indispensable; explanatory decompression asks what roles were combined and whether the data can distinguish them.

Newtonian gravitation illustrates the issue. In $F = Gm_1m_2/r^2$, the masses are sources and responses, r expresses spatial relation, inverse-square behavior reflects three-dimensional point-source spreading, and G fixes the coupling scale. Writing $\varepsilon_g = (4\pi G)^{-1}$ separates spherical normalization from inverse response without creating a new constant. The governing equation immediately re-couples these coordinates, so decomposition defines a constrained space rather than independent causes. The same discipline matters when a fitted effective coupling could represent geometry, response, source error, or a mixture.

1.2 A common family of problems

The examples are deliberately varied: circular phase closure, spherical point-source spreading, constrained transport, inertial continuation, galaxy dynamics, and cosmological sequences. They share one structure. A law usually admits a family of possibilities; geometry, source distribution, inherited state, boundaries, and present interaction jointly select a realized configuration. Historical claims add a further requirement: reconstructed states must be embeddable in at least one lawful sequence.

1.3 From privileged geometric forms to geometry-relative free motion

Greek astronomy and natural philosophy often treated the circle and sphere as privileged forms of perfection and regularity (Aristotle 1984). Modern physics retained their mathematical value while replacing perfection with symmetry, invariance, and model-dependent geometry. A circle expresses centered rotational symmetry; a sphere expresses isotropy about a point; a straight line expresses translational uniformity in Euclidean space. Relativity generalizes free continuation to geodesic motion, whose coordinate form depends on the metric. The historical shift motivates a methodological rule: geometric form should be read through the symmetry, constraint, and theory that generate it, not through a universal hierarchy of shapes.

1.4 Thesis and scope

$$R = \Phi_{\text{map}}(\Gamma, S, H, B, I)$$

Here Γ denotes spatial or spacetime geometry and directional structure; S denotes the physical source, its distribution and symmetry, and a declared reference source-field response scale; H denotes prior influence insofar as it is retained, attenuated, or transformed into the inherited present state; B denotes boundaries, media, matching conditions, and other admissibility constraints; I denotes the current interaction or evolution rule; and R denotes the realized physical configuration. The coordinates name explanatory roles, not independent ontological atoms or orthogonal mathematical variables. Governing equations immediately re-couple them.

Symbol note: R denotes the realized form in this role map and Φ_{map} denotes the explanatory mapping. They are distinct from the cylindrical radius R used in the galaxy calculations, the Ricci scalar R used in §4.8, and the gravitational potential Φ_{pot} used in Chapter 5.

$$(\Gamma_t, S_t, B_t) \rightarrow \mathcal{A}_t, (X_t, I_t; \mathcal{A}_t) \rightarrow X_{t+1} \rightarrow O_{t+1}$$

At is the admissible state or transition space and X_t is a dynamically sufficient inherited state at time t . Feedback is allowed: a later state may alter the source, geometry, or boundary conditions of the next step. In general relativity, for example, stress-energy contributes to the metric and the metric mediates free motion. The decomposition therefore preserves causal provenance without imposing a universal one-way ordering.

Explanatory decompression should be followed by restoration of the physical constraints, a data-likelihood rank audit, comparison of independent channels, a sequence test when history is invoked, and controlled recompression. Independent channels with nonparallel sensitivities can restore rank; in historical systems, lawful sequence linkage may further reduce epoch-specific freedom.

Epistemic boundary: Analytical separability is not empirical identifiability; empirical identifiability is not predictive success; predictive success is not a geometrical explanation; and none of these, by itself, establishes an ontology.

The examples move from phase closure, point-source spreading, constrained transport, inertia, and geodesic continuation to galaxy dynamics and cosmological sequences. The gravitational discussion is a weak-field research design, not a replacement for general relativity. The galaxy calculations are exploratory and post hoc relative to the literature used in model construction; black holes mark the boundary beyond which a scalar area or weak-field tensor cannot substitute for a covariant theory.

2. Physical Form as an Explanatory Target

2.1 A working definition of physical form

The phrase 'physical form' is used here in a deliberately limited sense. It does not invoke a Platonic realm of forms, nor does it reintroduce an Aristotelian substantial form as an independent ontological entity. It refers to the organized spatial, temporal, dynamical, or macroscopic manner in which a physical system is realized. A spherically symmetric electric field, a periodic oscillation, a quasi-one-dimensional current, a temperature gradient, uniform rectilinear motion, and geodesic free fall are all examples of physical form in this working sense.

The advantage of this notion is that it identifies an explanandum lying between ontology and numerical prediction. A law may tell us how quantities are related, but the observed phenomenon also has a shape, symmetry, dimensionality, temporal organization, and scale-dependent regime. Asking why a point-source field is spherical, why a thin conductor supports approximately one-dimensional transport, or

why one thermal boundary produces equilibration while another produces continuing heat loss is to ask why a lawful process is realized in a specific form.

2.2 Explanatory roles rather than independent first causes

The framework is intended to prevent premature causal compression. A variable can be important to a prediction without being an independent upstream cause. Geometry may be a structural condition, a mediator generated by a source, or part of the operator that defines propagation. A boundary may act as an admissibility constraint. Inherited state may mediate the influence of earlier interactions. An effective fitted coefficient may be a downstream proxy for geometry rather than a fundamental parameter. The relevant question is therefore not merely which quantity changes with the outcome, but what causal or explanatory authority that quantity possesses in the model.

Table 1. Explanatory roles and typical causal status in the structure-explicit framework.

Factor	Symbol	Primary explanatory role	Typical causal status in a model
Geometry	Γ	Dimension, metric, curvature, distance, directional equivalence, differential operators	Structural condition; mediator; propagation operator; sometimes an endogenous state
Source	S	What produces or carries the relevant physical quantity, with what symmetry or distribution, and the declared reference source–field response scale $\mathcal{E}_{g0,S}$	Source driver; parent variable; in GR a contributor to geometry through stress-energy
History / inherited state	H (static schema); X_t (dynamic form)	Selectively retained, attenuated, or transformed consequences of prior evolution; explicit finite or continuous memory only where required by the chosen reduced description; otherwise encoded in present state variables, ordinary propagators, retained constraints, or path dependence.	State mediator or inherited condition; carrier of explicit memory only when the model requires it; otherwise part of a dynamically sufficient present state; sequence node when ordered history matters.
Boundary / medium	B	Which configurations, fluxes, interfaces, and degrees of freedom are admissible	Constraint or moderator; sometimes an effective description of system–environment interaction
Current interaction	I	How the existing state changes under the active dynamical law	Transition driver; may be absent for free evolution or incorporated in a field equation
Realized form	R	The concrete field, motion, distribution, or macroscopic regime to be explained	Downstream observable or model realization; not a new primitive variable

2.3 Dynamic causal architecture and admissible state space

The static schema $R = \Phi_{\text{map}}(\Gamma, S, H, B, I)$ is useful for inventory, but causal interpretation benefits from a dynamic representation. Let $\mathcal{A}_t$ denote the states or transitions physically admissible at time t . Geometry, source structure, and boundaries may jointly restrict this space:

$$(\Gamma, S, B) \rightarrow \mathcal{A}_t.$$

Given that admissible structure, the inherited state and the current interaction determine which transition is realized: $(X_t, I_t; \mathcal{A}_t) \rightarrow X_{t+1} \rightarrow R_{t+1}$. The arrows express explanatory dependence, not a universal microphysical chronology. Feedback is allowed. A realized state can alter a source, a medium, or even the effective geometry used at the next step. The virtue of the diagram is that it separates role from interaction: coupled variables need not be collapsed into one undifferentiated coefficient.

2.4 Role-specific causal audit

Four questions discipline the use of the framework. First, a formal-admissibility question asks whether a proposed decomposition is algebraically, dimensionally, and theoretically legitimate. Second, a geometrical-generation question asks whether the specified geometry can actually generate the claimed change in the chosen observable, rather than merely correlate with it. Third, a sequential-historical question asks whether inherited state or an ordered pathway is necessary to obtain the realized form. Fourth, an empirical-residual question asks whether a deviation remains after geometry, source structure, boundaries, propagation, and measurement systematics have been modeled. This role-based audit is consistent with interventionist and structural approaches to causal explanation, while remaining deliberately less formal than a complete structural causal model (Pearl 2009; Woodward 2003).

The same audit also requires role classification. A high-impact quantity may be a direct driver, a mediator, a moderator, a proxy, a confounded indicator, or a reverse-causal signal. This distinction becomes decisive in Chapter 5: G_{eff} and $\varepsilon_{g,\text{eff}}$ can be fitted from data, but under the geometry-only hypothesis they are downstream recompressions of J_Γ and therefore do not automatically possess fundamental causal authority.

2.5 Background dimension and effective dimension

A central distinction concerns ambient dimension and the effective dimension of a process. A wire is a three-dimensional material body, but when its length greatly exceeds its transverse scale and the relevant observable is the macroscopic axial current, it can be modeled as a one-dimensional transport system. A thin film is embedded in three-dimensional space but may support approximately two-dimensional collective behavior. A particle constrained to a sphere has two positional degrees of freedom even though the sphere is embedded in three-dimensional Euclidean space.

Effective dimension is therefore not a denial of ambient dimension. It is a scale- and question-dependent characterization of the independent degrees of freedom that materially contribute to the phenomenon under study. This dependence on scale and observable is important: at microscopic resolution a wire contains three-dimensional lattice structure and random carrier motion; at macroscopic resolution the net transport may be accurately represented by one coordinate.

2.6 History as attenuated, retained, and transformed influence

History is not a second substance acting on the present. It is the selectively retained, attenuated, or transformed consequence of prior evolution encoded in a dynamically relevant state: momentum, phase, stress, composition, correlations, or a field configuration. A general audit notation is $H_t = R(H_{<t})$. Only a reduced model that genuinely requires explicit memory should introduce a kernel, for example

$$H(t) = \int K(t, s) Z(s) ds, \text{ or } H_t = \Sigma_r K_r Z_{t-r}$$

No universal decay, normalization, or sign is assumed. Generalized Langevin, Mori–Zwanzig, and master-equation formalisms show how eliminated variables can generate such terms (Lee 2000; Izvekov 2021; Vacchini 2016); weak-memory discrete dynamics also shows when a higher-order description can admit a controlled first-order reduction (Meyer and Brandner 2026). Conversely, enlarging the present state can make an evolution first-order without erasing the processes that formed that state.

2.7 Temporal Sequence Compatibility Principle

Historical retention and sequence compatibility are distinct. The first asks how earlier influence is represented now; the second asks which continuations remain admissible from that state:

$$H_{<t} -R \rightarrow H_t \rightarrow X_t -T_t \rightarrow X_{t+1}$$

$$(X_{t-\Delta t}, X_t, X_{t+\Delta t}) \in \mathcal{C}_t(\Gamma_t, S_t, B_t, I_t)$$

$\mathcal{C}_t$ may encode a deterministic map, stochastic support, a higher-order equation represented on an augmented state, or inequality constraints. The principle does not assert discreteness, determinism, or a universal arrow of time. It requires only that a multi-time explanation place its states within a history admitted by the declared model. It is a necessary filter on historical narratives, not a complete causal theory.

2.8 Boundary conditions as constitutive constraints

Boundary conditions are often presented as auxiliary data appended after the law. In many applications, however, they help constitute the phenomenon. Without the fixed length and pivot of an ideal pendulum there is no pendulum system; without a conducting channel there is no corresponding quasi-one-dimensional current; without a thermal interface that fixes temperature or suppresses heat flux

there is no determinate heat-conduction problem. A boundary can therefore act by restricting the admissible state and flux space rather than by supplying an additional force.

This does not make boundaries metaphysically fundamental. They may be effective descriptions of more detailed interactions between system and environment. At the level at which a model is posed, however, they can be indispensable to mathematical well-posedness, causal interpretation, and the realized form of the solution.

A short diffusion calibration belongs here rather than in a separate case chapter. For $\partial_t T = \alpha \nabla^2 T$, the diffusivity α does not by itself select a temperature field: geometry, initial data, and boundary conditions determine the realized solution (Carslaw and Jaeger 1959). This familiar example calibrates the role map without supporting a new claim about gravity.

3. Circle, Sphere, and Line: Distinct Forms of Geometric Uniformity

3.1 Symmetry and dimensional role

The circle, sphere, and line express different invariances. A circle is uniform under rotations about a center; a sphere is isotropic under three-dimensional rotations about a point; a Euclidean line is homogeneous under translation. These claims concern relational structure, not a metaphysical ranking of shapes. The same curve can also occupy several dimensional descriptions: a circle is embedded in a two-dimensional plane, has a one-dimensional constrained configuration space S^1 , and may be sampled through a still lower-dimensional observable. Embedding dimension, intrinsic dimension, constraint dimension, and effective transport dimension must therefore be stated separately.

3.2 Limits and approximations

A straight line can arise as the zero-curvature limit of a circle because $\kappa = 1/R \rightarrow 0$ as $R \rightarrow \infty$. It can also arise locally from a smooth curve or effectively when boundaries suppress transverse degrees of freedom. These are distinct mechanisms and should not be collapsed into the claim that every line is literally a circle of infinite radius. Likewise, projection or coarse-graining may hide variation without changing the underlying geometry. The methodological lesson is to identify which dimension, symmetry, and limiting operation support the form used in an equation.

4. Geometric Factors in Physical Equations: 2π , 4π , and 8π

4.1 2π and phase closure

A planar pendulum is embedded in a two-dimensional vertical plane, but the fixed-length constraint $x^2 + y^2 = L^2$ reduces its admissible configuration to the one-dimensional manifold S^1 . Ordinary oscillation realizes only an arc, not a complete spatial circle. In the small-angle limit, however, its phase advances periodically:

$$\ddot{\theta} + (g/L)\theta = 0, \quad T = 2\pi\sqrt{L/g}$$

The factor 2π therefore reflects genuine circular structure, but the relevant circle is the phase circle S^1 , whose complete cycle has angular measure 2π radians. Embedding space, constrained configuration space, realized path, and phase representation should not be conflated.

4.2 4π , solid angle, and inverse-square spreading

In three-dimensional Euclidean space the area of a sphere of radius r is $4\pi r^2$. If an isotropic point source generates a conserved outward flux Φ_{flux} , then the flux density on each concentric sphere obeys

$$\Phi_{\text{flux}} = E(r) 4\pi r^2 = \text{constant}$$

$$E(r) \propto \frac{1}{r^2}$$

The inverse-square form therefore depends on more than the intrinsic nature of the source. It depends on point-source idealization, three-dimensional isotropic spreading, and conservation of flux. If the source is extended, if the spatial dimension differs, if the medium absorbs, or if boundaries break symmetry, the distance dependence may change.

The dimensional generalization makes the point more clearly than numerical comparison among 2π , 4π , and 8π . In d -dimensional Euclidean space, conserved point-source flux crosses a $(d-1)$ -sphere whose unit area is

$$\Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}$$

Here $\Gamma(d/2)$ is the Euler gamma function; it is unrelated to the geometry-role label Γ used elsewhere in the framework.

and the corresponding flux density scales as $r^{-(d-1)}$. The stable structural principle is not a doubling sequence of π -factors but conservation divided by the growth of equidistant surfaces.

4.3 Geometry and coupling scale

A physical coefficient often combines geometrical normalization with a coupling scale. In Coulomb's law, 4π expresses spherical normalization while ϵ_0 participates in the chosen representation of electromagnetic coupling. In Newtonian gravitation, the inverse-square form reflects spherical spreading while G sets the strength with which mass sources gravitational effects. The heuristic distinction can be expressed as follows: geometry primarily constrains the form of distribution; coupling constants set the scale of interaction. Yet standard notation may package these roles together. A structure-explicit reformulation can separate them for explanatory purposes without claiming that the separated factors are independent physical entities. The distinction is also not absolute, because constants such as c and $\hbar$ organize the structure of whole theories rather than merely multiplying a fixed law.

4.4 From $4\pi G$ to $8\pi G/c^4$

General relativity replaces the Newtonian scalar relation by $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ (Einstein 1915, 1916; Wald 1984). The factor 8π is not the area of a four-dimensional sphere and its factor of two does not encode the addition of time. Its normalization follows from the Einstein tensor, the contracted Bianchi identity, conventions, and recovery of Newtonian gravity in the weak-field limit. Numerical factors acquire meaning from their equations and limiting relations, not from free geometrical analogy.

4.5 Compact notation and explanatory decompression

Standard physical equations are optimized for calculation, comparison, and communication. Their compactness is indispensable, but it can obscure the distinct origins of their terms. A single coefficient may incorporate a geometrical normalization and a coupling strength; a simple point-particle law may suppress the field equation from which it is derived; and initial or boundary conditions may be treated as external data even though they help determine the realized phenomenon. This paper calls the recovery of these roles explanatory decompression.

Explanatory decompression is not a proposal to replace standard notation. It is a supplementary representation in which geometrical factors, source terms, response variables, coupling scales, inherited states, and boundary assumptions are made explicit when doing so clarifies their explanatory functions. The criterion of success is not algebraic novelty but improved discrimination: the reformulation should show which features are geometrical, which are dynamical, which are constitutive, and which are idealized or suppressed.

Standard equations compress; structural analysis decompresses.

4.6 The gravitational response parameter epsilon-g (ϵ_g)

$$\epsilon_g \equiv 1/(4\pi G), \quad G = 1/(4\pi\epsilon_g)$$

This is a reparameterization, not a new empirical constant. It lets the point-mass field and local source equation display different roles:

$$g(r) = -M \hat{r}/(4\pi\epsilon_g r^2), \quad \nabla \cdot g = -\rho_m/\epsilon_g$$

The auxiliary flux $D_g = \epsilon_g g$ then satisfies $\nabla \cdot D_g = -\rho_m$. This notation exposes the inverse source–field response scale but supplies no mechanical vacuum ontology. Its value contains exactly the information already contained in G .

4.7 Electromagnetic analogy: a heuristic, not an identity

The epsilon-g notation makes a genuine formal analogy visible:

$$\nabla \cdot E = \rho_q/\epsilon_0, \quad \nabla \cdot g = -\rho_m/\epsilon_g.$$

Both equations exhibit a source–flux–response structure, and both point-source fields acquire an inverse-square form under isotropic propagation in three spatial dimensions. This similarity is scientifically useful because it identifies common mathematical organization and motivates comparison with weak-field gravitoelectromagnetism. It does not show that gravitation is simply electromagnetism with mass substituted for charge.

The differences are decisive. Electromagnetism couples to electric four-current and is represented by a vector gauge field; general relativity couples universally to the stress–energy tensor and is represented by metric and curvature tensors. Electric charge has opposite signs and permits attraction or repulsion, whereas ordinary positive mass–energy produces attractive Newtonian gravity. The gravitational field also contributes to gravitational dynamics through nonlinear self-coupling. Structural analogy can therefore guide inquiry without establishing ontological identity or proving the existence of a particular exchange particle.

4.8 epsilon-g in Einstein’s equation and the curvature-response scale

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 2T_{\mu\nu}/(\epsilon_g c^4), \quad K_g \equiv \epsilon_g c^4/2$$

$K_g = c^4/(8\pi G)$ is the inverse curvature-response coefficient associated with the standard normalization. Calling it a spacetime stiffness is only an analogy. The same constant weights the Einstein–Hilbert curvature term, $S_g = (\epsilon_g c^3/4) \int (R - 2\Lambda) \sqrt{-g} d^4x$. None of these rewritings adds dynamics; it displays the common coupling at Newtonian and relativistic levels. Here R denotes the Ricci scalar and g the metric determinant; neither symbol denotes the realized form R or the gravitational field g used in the weak-field chapters.

4.9 Beyond the sphere: geometrical normalization and gravitational response

After $\varepsilon_g = (4\pi G)^{-1}$ is substituted, the explicit π disappears from Einstein's equation algebraically; general relativity has not escaped from a sphere. In the Newtonian point-source solution, $4\pi r^2$ belongs to a symmetry-reduced flux surface, whereas ε_g remains in the local source–field relation for nonspherical configurations. In general relativity no spherical radius need appear at all: geometry and source are tensorially coupled, and spherical symmetry characterizes particular solutions. Thus ε_g is a useful response coordinate, not an ontologically more fundamental constant.

5. Geometry-Dependent Gravitational Response: Constraints, Recompression, and Residual Tests

This chapter instantiates the preceding roles rather than adding a second framework. In the weak-field numerical case, the general structural role Γ_t is represented by $T_R(t, x) = J_R \mathcal{L}_R[\beta_t]$, while historical inheritance H_t is carried by an augmented state $X_t^{\text{aug}} = (X_t^{\text{matter}}, \beta_t)$. Because the actual baryonic distribution already enters ρ_S , T_R is initially a neutral residual representation; it may be promoted to T_Γ only through an independent metric or geometrical gate. The closure $\tau_\beta \partial_t \beta = \beta_{\text{target}}[\rho_S, Q_{\text{flat}}] - \beta + \tau_\beta D_\beta \nabla^2 \beta$ makes the retained response state explicit. Eliminating β yields an exponentially weighted memory, whereas retaining it gives a first-order Markov representation on the augmented state. The finite-volume experiment sets $J_R = 1$ and $\mathcal{L}_R = \text{diag}(e^\beta, e^\beta, e^{-2\beta})$; it tests one declared Γ –H coupling, not the claim that galaxy spacetime is literally a lattice or that this closure is unique.

5.1 The S– Γ decomposition and a structure-explicit baseline

$$\varepsilon_{g0,S} \equiv 1/(4\pi G_0), \quad G_0 = 1/(4\pi \varepsilon_{g0,S})$$

This definition introduces no empirically independent constant. Measuring $\varepsilon_{g0,S}$ is measuring the reference Newtonian coupling in another parameterization. The subscript S records its explanatory assignment as the reference source-to-field response scale, whereas $4\pi r^2$, A_Γ , directionality, and metric or differential structure are assigned to Γ . The point of the notation is to stop the conventional symbol G from prematurely recompressing these roles. It is an explanatory classification, not a claim that response can vary independently of geometry in a completed physical theory.

$$g_\Gamma(r) = -M_S(r)/[\varepsilon_{g0,S} A_\Gamma(r)]$$

A_Γ is an observable-specific response measure derived for a stated geometry, symmetry class, and approximation. It is a literal area only when the theory derives it as the relevant equidistant flux surface. In a general relativistic setting, lapse, redshift, connection, causal propagation, spin, and the Green operator may all matter; a single area ratio is not a universal curvature scalar.

Role assignment: $S = \{M_S, \rho_S, \varepsilon_{g0,S}, \text{source distribution}\}$; $\Gamma = \{A_\Gamma, J_\Gamma, \text{directionality, metric and tensor operators}\}$. **The governing field equation must restore their physical coupling.**

5.2 Three representations and a same-dimensional reparameterization

$$J_\Gamma(r) = A_\Gamma(r)/(4\pi r^2), \quad \varepsilon_{g,\text{eff}} = \varepsilon_{g0,S} J_\Gamma, \quad G_{\text{eff}} = G_0/J_\Gamma$$

On the geometry-only branch with the S-role reference fixed, $\varepsilon_{g,\text{eff}}/\varepsilon_{g0,S} = J_\Gamma = G_0/G_{\text{eff}}$. These are alternative scalar recompressions of one structure-explicit relation, not a new law. If both geometry and response are allowed to vary, $G_{\text{eff}} = G(\varepsilon_g)/J_\Gamma$ is many-to-one. A successful fit in that compressed coordinate cannot decide which decompressed role produced it.

5.3 From scalar area to a residual tensor lattice

A scalar J can encode overall spreading but not the distribution of flux among directions at the same radius. Introduce a gravitational-flux vector D_g and a dimensionless second-rank response lattice T_R :

$$D_g = \varepsilon_{g0,S} T_R g, \quad \nabla \cdot D_g = -\rho_S, \quad g = -\nabla \Phi_{\text{pot}}$$

$$\nabla_i [\varepsilon_{g0,S} T_R^{ij} \nabla_j \Phi_{\text{pot}}] = \rho_S$$

The field g remains a vector; the tensor maps field to flux. This is a static or quasi-static weak-field Poisson representation. It is neither Einstein's equation nor a derivation from a spacetime metric. A useful normalization is

$$T_R = J_R \mathcal{L}_R, \quad \det \mathcal{L}_R = 1$$

J_R retains the scalar response scale while $\mathcal{L}_R$ carries only shape and direction. The isotropic reference is $J_R = 1$ and $\mathcal{L}_R = \mathbf{I}$. An axisymmetric toy lattice may be parameterized as

$$\mathcal{L}_R(\beta) = \text{diag}(e^\beta, e^\beta, e^{-2\beta})$$

For a constant-coefficient point-source toy model, equipotentials satisfy $x^T \mathcal{L}_R^{-1} x = \text{constant}$, the equal-radius equatorial-to-polar field ratio is $g_{\text{eq}}/g_{\text{pole}} = e^{3\beta/2}$, and a spherical equatorial inversion gives $M_{\text{iso,eq}}/M_{\text{true}} = e^{\beta/2}/J_R$. These relations are conditional and falsifiable; they are not descriptions of an observed galactic spatial metric. The empirical object remains T_R or J_R . Promotion to T_Γ or J_Γ is allowed only when an independent metric theory, structural derivation, or geometrical measurement supplies the mapping.

Within this declared constant-coefficient model, the ellipsoidal equipotentials are not inserted as an independent geometrical assumption; they follow analytically from the anisotropic Poisson operator and the point-source boundary problem. The constitutive assumption lies instead in the choice of $\mathcal{L}_R(\beta)$ and, in

the dynamic case, in the closure governing its evolution. Whether the resulting response corresponds to a physical spatial metric remains a separate empirical question.

No double counting: A flattened or irregular source already produces an anisotropic Newtonian field in flat space. If a lattice is fitted to that same field, it is a low-dimensional representation. Only a residual left after the full baryonic source geometry has been modeled can be a candidate for new geometrical interpretation.

5.4 From a response tensor to a geometrical tensor

Suppose that an independently specified positive-definite spatial metric h_{ij} governs the weak-field propagation of a scalar potential. Writing

$$h = \det(h_{ij}),$$

the corresponding Laplace--Beltrami operator is

$$\Delta_h \Phi_{\text{pot}} = \frac{1}{\sqrt{h}} \partial_i (\sqrt{h} h^{ij} \partial_j \Phi_{\text{pot}}).$$

If ρ_{phys} denotes mass density per unit proper volume, the metric field equation

$$\Delta_h \Phi_{\text{pot}} = \frac{\rho_{\text{phys}}}{\varepsilon_{g0,S}}$$

is equivalently written as

$$\partial_i [\varepsilon_{g0,S} \sqrt{h} h^{ij} \partial_j \Phi_{\text{pot}}] = \sqrt{h} \rho_{\text{phys}}.$$

It is therefore useful to define

$$T_{\Gamma}^{ij} \equiv \sqrt{h} h^{ij}, \quad \rho_{\text{coord}} \equiv \sqrt{h} \rho_{\text{phys}},$$

so that

$$\partial_i [\varepsilon_{g0,S} T_{\Gamma}^{ij} \partial_j \Phi_{\text{pot}}] = \rho_{\text{coord}}.$$

Strictly, T_{Γ}^{ij} is a contravariant tensor density of weight +1, rather than an ordinary rank-two tensor. Likewise, ρ_{coord} is the source density relative to the coordinate volume element d^3x . Their pairing preserves the physical source measure,

$$M_S = \int \rho_{\text{phys}} \sqrt{h} d^3x = \int \rho_{\text{coord}} d^3x.$$

In a fixed coordinate chart and in three positive-definite spatial dimensions, the algebraic relation can be inverted:

$$h_{ij} = \det(T_{\Gamma}) (T_{\Gamma}^{-1})_{ij}.$$

This inversion is a coordinate-level reconstruction, not by itself evidence that a phenomenological response tensor is a physical spatial metric. Identifying the weak-field T_R^{ij} with the metric-derived T_Γ^{ij} requires a declared coordinate and volume measure, a compatible mapping between ρ_S and ρ_{phys} , and an independently specified metric theory. The resulting structure must also make consistent predictions for matter dynamics and light propagation. Moreover, a constant anisotropic operator may represent only a coordinate rescaling of flat space; genuine curvature requires position dependence together with nonzero Riemann curvature or nontrivial holonomy.

Geometrical promotion gate: $T_{R,\text{phen}}$ + specified volume-measure and source-density mapping + independent metric derivation + lensing/off-plane consistency $\Rightarrow$ possible identification with T_Γ . A successful fit or algebraic inversion alone does not complete this inference.

Geometrical promotion gate: $T_{R,\text{phen}}$ + independent metric derivation + lensing/off-plane consistency $\Rightarrow$ possible T_Γ . A successful fit alone does not complete this inference.

5.5 Constraint manifold, implicit coupling, and closure

$$F(\Phi_{\text{pot}}, T_R, \rho_S) = \nabla \cdot (T_R \nabla \Phi_{\text{pot}}) - \rho_S / \varepsilon_{g0,S} = 0$$

$$\nabla \cdot (\delta T_R \nabla \Phi_{\text{pot}} + T_R \nabla \delta \Phi_{\text{pot}}) = \delta \rho_S / \varepsilon_{g0,S}$$

In the isotropic Euclidean limit, let $g_{\text{mag}} = |\nabla \Phi_{\text{pot}}|$, $n = \nabla \Phi_{\text{pot}} / g_{\text{mag}}$, and $K_\Sigma = \nabla \cdot n$, where K_Σ is the extrinsic mean-curvature trace of an equipotential level set.

$$\rho_S = \varepsilon_{g0,S} (\partial_n g_{\text{mag}} + K_\Sigma g_{\text{mag}})$$

This normal decomposition of the Poisson equation makes source density, the normal variation of field strength, and equipotential curvature mutually constrained. It does not provide the material-evolution closure, and it does not identify equipotential curvature K_Σ with spatial or spacetime curvature. In the tensor case, directional projections and gradients of T_R contribute additional terms.

These relations show that field, source, and response are mutually constrained under flux conservation. The field equation tells us how to solve for Φ_{pot} when T_R and ρ_S are specified, but it does not explain why T_R evolves as it does. A genuinely co-generative model must therefore declare an independent closure $C(T_R, \Phi_{\text{pot}}, \rho_S, H) = 0$. Inferring T_R from an outcome and then reusing it as the cause of that same outcome without a closure would be circular.

$$F(g, \varepsilon_{g,S}, A_\Gamma; M_S) = g \varepsilon_{g,S} A_\Gamma - M_S = 0$$

In the scalar symmetry limit and at fixed M_S , the positive solutions form a regular two-dimensional constraint manifold. The implicit-function theorem permits any one coordinate to be expressed locally in terms of the other two. Analytical decomposition therefore does not create a free Cartesian ontology; the

physical equation immediately restores the mutual constraint among field, source-response scale, and structural response.

$$d \ln g = d \ln M_S - d \ln A_r - d \ln \varepsilon_{g,S}$$

At fixed source mass, a change in one scalar field measurement does not determine whether Γ changed, the S-role response changed, or both changed. Increasing the precision of the same scalar channel cannot remove this structural null direction.

5.6 Likelihood rank, priors, and independent channels

$$J_{\text{data}} = \partial O / \partial \theta$$

Structural identifiability is first a property of the forward map and the data likelihood. If $J_{\text{data}} = (j_S, j_R, j_\eta)$, separate identification of the S and residual-response directions requires the relevant column to leave the column space generated by the others. Informative priors can make a posterior finite and stabilize a numerical estimate, but they do not constitute empirical rank supplied by an observational channel. The reporting order should therefore be: data Jacobian or likelihood Fisher information, singular values and condition number, profile likelihood or equivalent data-only diagnostics, and only then prior-conditioned practical identifiability (Raue et al. 2009).

Reporting rule: Informative priors may regularize estimation but must not be counted as empirical rank supplied by the observational channel.

5.7 Weak-field flux conservation and directional redistribution

$$\nabla \cdot (\varepsilon_{g0,S} T_R g) = -\rho_S, \quad \oint \varepsilon_{g0,S} T_R g \cdot dA = -M_S$$

Flux conservation fixes the integral through a closed surface, not equal field strength in every direction. A flattened response can enhance the equatorial sector only if other directions compensate. Define the normalized axisymmetric angular flux weight as

$$w(\theta) = 1 + a_2 P_2(\cos \theta) + a_4 P_4(\cos \theta), \quad \langle w \rangle_\Omega = \frac{1}{4\pi} \int_{4\pi} w(\theta) d\Omega = 1.$$

Because the higher Legendre modes have zero spherical mean, flux conservation fixes the monopole coefficient to unity, while a_2 and a_4 redistribute the conserved flux over direction. Equatorial data constrain

$$w_{\text{eq}} - 1 = -\frac{a_2}{2} + \frac{3a_4}{8},$$

whereas one polar channel constrains

$$w_{\text{pole}} - 1 = a_2 + a_4.$$

The determinant of these two sensitivity directions is $-7/8$, so combining equatorial and polar observations changes the angular problem from rank one to rank two.

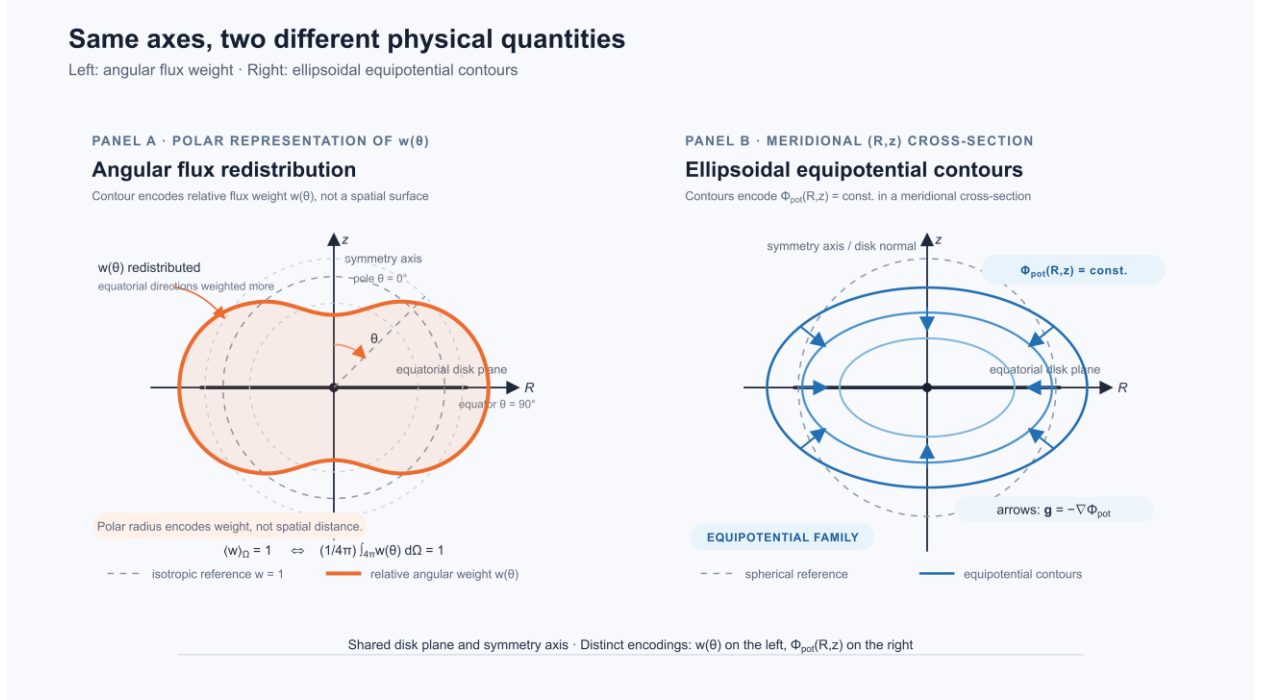


Figure 1. Angular flux redistribution and ellipsoidal equipotential contours as distinct representations. Panel A shows the dimensionless angular flux weight $w(\theta)$, normalized by $\langle w \rangle_\Omega = 1$. Its polar radius encodes relative directional weight rather than spatial distance. Panel B shows a meridional (R, z) cross-section of a model ellipsoidal equipotential family, $\Phi_{\text{pot}}(R, z) = \text{const.}$; the arrows indicate the local field direction $\mathbf{g} = -\nabla \Phi_{\text{pot}}$. The two panels share the same disk plane and symmetry axis but represent different physical quantities. The dashed curves provide their respective isotropic and spherical references; ellipsoidal equipotentials do not by themselves establish an ellipsoidal spatial metric.

Conservation removes an arbitrary monopole freedom but does not by itself add empirical rank. With equatorial rotation curves alone, mass normalization, $\varepsilon_{g0,S}$, an isotropic halo contribution, and a directional response can remain confounded. Vertical dynamics, polar rings, stellar streams, satellite motions, lensing, or independently calibrated source mass can supply nonparallel directions. Conservation constrains admissibility; observations restore data rank.

5.8 A flux-conserving, history-coupled generative experiment

A static inversion can show that some T_R rewrites an observed field, but it cannot show that the response and source can co-generate a continuous history under conservation. To convert the implicit relation into an executable conditional model, an axisymmetric weak-field finite-volume experiment was constructed. Matter is represented by axisymmetric mass rings with fixed specific angular momentum, radial and vertical damping, and pressure regularization. The same initial particle state is evolved in a

Newtonian branch with $T_R = \mathbf{I}$ and in a history-coupled branch. This is a methodological audit, not a high-fidelity simulation of galaxy formation.

$$\nabla \cdot (T_R \nabla \Phi_{\text{pot}}) = \rho_S / \varepsilon_{g0,S} = 4\pi G_0 \rho_S, \quad T_R = \text{diag}(e^\beta, e^\beta, e^{-2\beta})$$

$$Q_{\text{flat}} = [\langle R^2 \rangle - 2\langle z^2 \rangle] / [\langle R^2 \rangle + 2\langle z^2 \rangle]$$

$$\tau_\beta \partial_t \beta = \beta_{\text{target}}[\rho_S, Q_{\text{flat}}] - \beta + \tau_\beta D_\beta \nabla^2 \beta$$

The condition $\det T_R = 1$ fixes the local volume normalization of the shape tensor, while the finite-volume discretization makes internal face fluxes cancel pairwise. The target β is built from a smoothed density weight and the global flattening Q_{flat} ; β relaxes on a finite timescale τ_β and diffuses, thereby representing explicit historical inheritance. The implementation uses dimensionless units with $G_0 = 1$ and hence $\varepsilon_{g0,S} = 1/(4\pi)$. The code checks that the unscaled boundary lattice flux equals $4\pi G_0 M$ and that the normalized D_g flux equals M .

$$\rho_{S,t} \rightarrow \beta_{\text{target},t} \rightarrow T_{R,t} \rightarrow \Phi_{\text{pot},t} \rightarrow X_{\text{matter},t+1} \rightarrow \rho_{S,t+1}$$

This is the feedback loop actually implemented by the finite-time generator. It does not prescribe an ellipsoidal boundary at each step; nonspherical matter and response arise conditionally from the declared density–field–response closure. Because β_{target} is chosen rather than derived from a metric theory or established microphysics, and because the late-time audit does not establish equilibrium, the generated morphology remains conditional rather than a prediction of galactic spatial geometry.

Generative structure: The field equation $\mathbf{F} = 0$ supplies the instantaneous conservation constraint; the closure $\mathbf{C} = 0$ supplies the historical evolution of T_R . Without $\mathbf{F}$ the model is not conservative, and without $\mathbf{C}$ it is only a post hoc inversion rather than a generative explanation.

5.9 Declared finite-time run: invariant total flux, altered local inversion

The declared main run used a 56×72 grid, 6,000 mass rings, $\Delta t = 0.02$, and 280 steps, ending at the finite dimensionless time $t = 5.6$. The two branches share the same initial state, mass, angular momentum, numerical regularization, and boundary conditions; only the history-coupled T_R differs. Because the late-time audit below does not satisfy the quasi-steady criterion, $t = 5.6$ is an evaluation time rather than an equilibrium or asymptotic endpoint.

Table 2. Conditional comparison at $t = 5.6$ for two runs with an identical initial state

Diagnostic at $t = 5.6$	Newtonian	History-coupled	Coupled relative to baseline
Scale height	0.14491	0.10455	−27.85%
Spherical-equivalent apparent mass at $R = 3$	1.86630	2.17872	+16.74%
Peak-density radius	2.30357	2.19643	−4.65%

Diagnostic at $t = 5.6$	Newtonian	History-coupled	Coupled relative to baseline
Central radial curvature $\mathcal{K}_R$	-0.03797	-0.08208	$ \mathcal{K}_R \approx 2.16\times$
Central vertical curvature $\mathcal{K}_Z$	0.07295	0.27036	$\approx 3.71\times$
Mass-weighted β	0	0.42557	Closure state

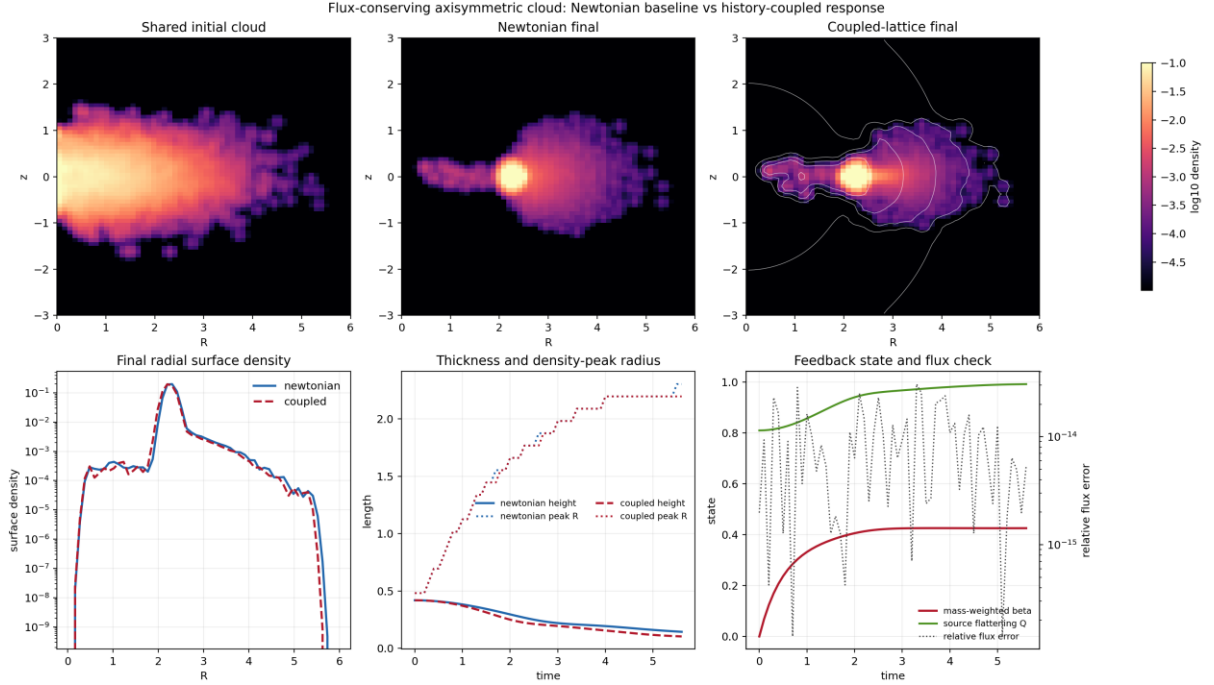


Figure 2. Generative evolution in the flux-conserving axisymmetric cloud model. The upper row shows the common initial state, the Newtonian final state, and the history-coupled final state; the lower row shows radial surface density, scale height/peak-density radius, and β –source flattening–flux diagnostics. The differences are consequences of the declared closure, not observations of a galactic spatial metric with ellipsoidal level sets.

With total mass and boundary flux held fixed, the coupled branch develops stronger vertical focusing and a larger spherical-equivalent apparent mass. The experiment therefore gives a direct numerical example of the distinction total-flux conservation $\neq$ local isotropy $\neq$ invariant inferred mass. The maximum relative flux error in the main run is 3.07×10^{-14} . The peak-density radius shifts inward by roughly one grid cell, but this diagnostic is not sufficiently stable to serve as a primary prediction. The tabulated amplitudes are conditional on the declared domain and closure and are superseded, for quantitative interpretation, by the domain and time-horizon audit below.

5.10 Systematic and remaining convergence audits

To avoid treating one visually attractive trajectory as a robust discovery, sixteen configurations were run to the same final time $t = 2.8$. The grid family ranges from 40×52 to 68×88 ; time steps are 0.04, 0.02, and 0.01; initial vertical scales correspond to $Q_{\text{flat}} \approx 0.411$ –0.911; and five closures—linear density, global linear, saturating, quadratic, and instantaneous—are compared. All smoothing and diffusion lengths are held fixed in physical grid units.

Table 3. Systematic scan and numerical-uncertainty summary

Scan dimension	Comparison	Numerical result	Interpretation
Grid convergence	56×72 vs 68×88	Height 0.62%; apparent mass 1.05%; $\mathcal{K}_R$ 0.88%; $\mathcal{K}_Z$ 1.63%	Key effects are converged to approximately the 1% level
Time-step convergence	$\Delta t = 0.02$ vs 0.01	Height 0.12%; apparent mass 0.07%; $\mathcal{K}_R$ 0.14%; $\mathcal{K}_Z$ 0.48%	Time-discretization error is smaller than model-form variation
Initial flattening	$Q \approx 0.411 \rightarrow 0.911$	Height effect $-3.30\% \rightarrow -18.55\%$; mass effect $+9.30\% \rightarrow +21.62\%$	Flatter initial states generally strengthen the selected feedback
Closure relation	Five forms	Height effect -11.36% to -15.57% ; mass effect $+17.63\%$ to $+26.92\%$	Sign is stable; amplitude has model-form uncertainty
Flux closure	All configurations 16	Maximum relative error 1.65×10^{-13}	Discrete non-conservation does not produce the differences

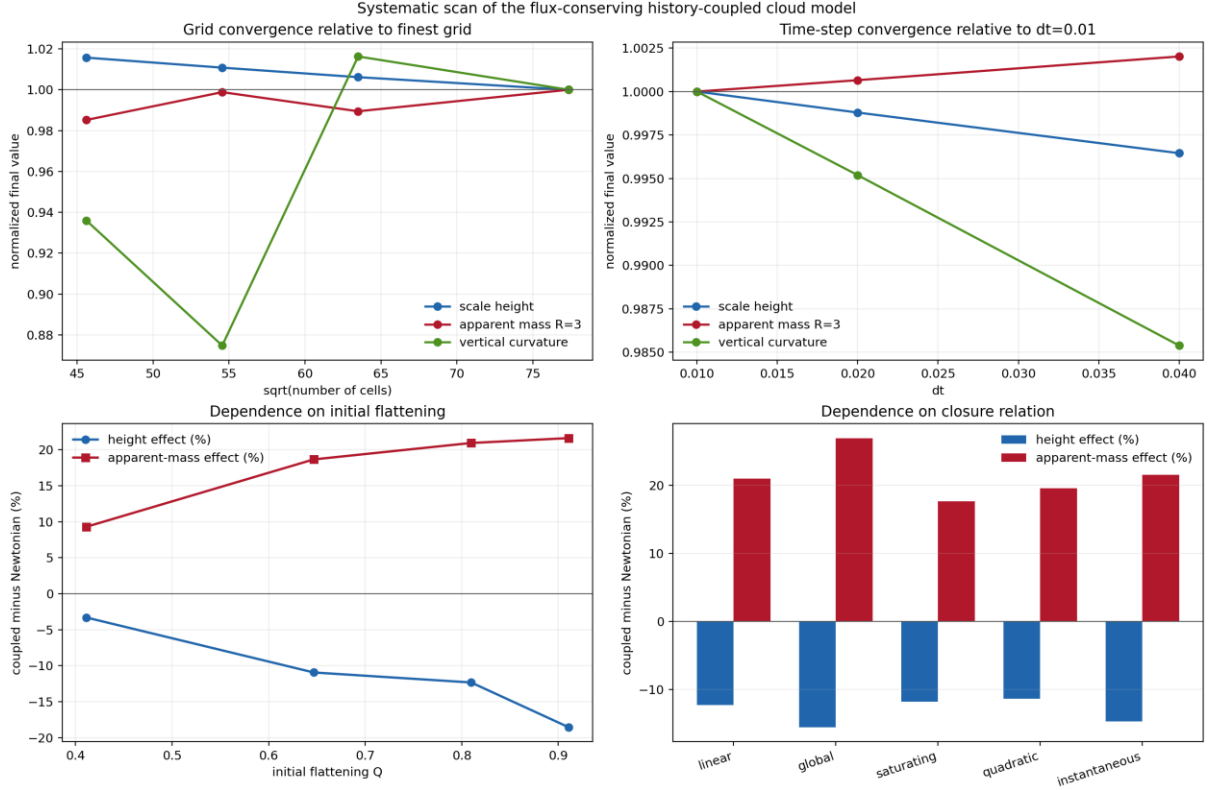


Figure 3. Systematic scan of the flux-conserving history-coupled model. The upper panels test grid and time-step convergence; the lower panels show how initial flattening and closure form change the coupled-minus-Newtonian effects.

The remaining audit added particle counts of 3,000, 6,000, and 12,000; five fixed random seeds; four successively expanded domains; three regularization lengths; linear-solver residuals; particle Courant numbers; boundary occupancy; and a late-time drift test. At 6,000 versus 12,000 particles, the relative differences are 3.40% for the height effect and 1.31% for the apparent-mass effect. Across seeds, the corresponding coefficients of variation are 5.29% and 0.83%. Both tests pass their declared tolerances. The maximum solver residual is 2.071×10^{-15} , the maximum particle Courant number is 0.179, and no particles occupy the monitored boundary region.

The finite-domain result is more restrictive. Relative to the largest tested domain, the original box differs by 25.18% in the height effect and 12.78% in the apparent-mass effect and therefore fails the declared 10% adequacy criterion. The last two expanded domains, however, differ by only 1.23% and 0.06%; at $t = 2.8$ the largest domain gives a conditional height effect of -16.48% and an apparent-mass effect of $+24.01\%$. These values are the better-resolved finite-time amplitudes for the declared closure, not estimates of a real galactic mass component.

The late-time interval also remains dynamic: from $t = 4.2$ to 5.6 the height and apparent-mass effects change by 23.15% and 18.16%, respectively, failing the declared 5% quasi-steady criterion. All endpoint runs retain the qualitative sign of a thinner coupled branch and a larger spherical-equivalent apparent mass, but neither the $t = 2.8$ nor the $t = 5.6$ configuration should be described as an equilibrium or

asymptotic prediction. The audits therefore separate robust sign, conditionally converged amplitude, and unresolved long-time behavior.

Convergence boundary: Numerical convergence establishes that the reported sign is not readily attributable to the tested discretization, sampling, or solver errors. It does not validate the closure in nature, supply empirical rank, or establish the physical origin of the generated effect.

5.11 Methodological contribution of the generative simulation

The experiment moves the paper from a static parameter rewrite to a constrained generative test. Separating $\varepsilon_{g0,S}$ and T_R is no longer merely a notational preference: once both are compressed into one G_{eff} , total flux, angular compensation, the historical lag of β , and source evolution cannot be audited separately. The explicit decomposition creates distinct residuals for conservation, convergence, closure robustness, held-out prediction, and cross-direction observation.

It also generates an observational design. If a real residual represents directional redistribution, equatorial enhancement should be accompanied by off-plane compensation and may covary with vertical dynamics, disk thickness, polar rings, stream precession, weak-lensing quadrupoles, or the evolution of source flattening. Adding more rotation curves in the same direction need not break the degeneracy among mass, $\varepsilon_{g0,S}$, isotropic extra mass, and T_R .

Evidential status: The current result is a model-existence result: the declared closure can generate thinner structure and a larger Newtonian-equivalent apparent mass while conserving total flux. It is not evidence that nature uses this closure, and it is not evidence that dark matter is absent.

5.12 Numerical comparison with the Newtonian exponential-disk field

To determine whether a tensor lattice adds new physics or merely compresses known source geometry, an axisymmetric infinitesimally thin Newtonian exponential disk, $\Sigma(R) = M \exp(-R/R_d)/(2\pi R_d^2)$, was integrated over angle. The exact Newtonian angular flux on each spherical shell was expanded in Legendre modes. The monopole was fixed by the enclosed mass and flux conservation; $l = 2$ and $l = 4$ terms were then fitted to directional structure.

$$q_{\text{disk}}(\mu; r) = -g_r(r, \mu)r^2/(G_0 M) = -4\pi\varepsilon_{g0,S} g_r(r, \mu)r^2/M$$

$$g_r^l(r, \theta) = -(G_0 M/r^2) \Sigma_l a_l(r) P_l(\cos \theta)$$

$$\oint g \cdot dA = -4\pi G_0 M a_0(r) = -M a_0(r)/\varepsilon_{g0,S}$$

These expressions preserve the paper's notation $G_0 = 1/(4\pi\varepsilon_{g0,S})$. The dimensionless coefficients a_l carry the angular representation; $\varepsilon_{g0,S}$ remains the inverse source-response coordinate and is not silently redefined as G_0 . For the Newtonian control, $a_0(r)$ equals the baryonic mass fraction enclosed by the spherical shell, while a_2 , a_4 , and higher modes redistribute the same total flux over angle.

Table 4. Angular comparison between the Newtonian exponential disk and low-order lattice compressions

r/R_d	Newtonian $g_{\text{eq}}/g_{\text{pole}}$	$l = 0 + 2$ angular RMS	$l = 0 + 2 + 4$ angular RMS
1.0	1.133	0.603%	0.166%
2.2	1.666	5.908%	3.269%
3.0	1.722	6.432%	3.602%
6.0	1.396	3.235%	1.654%
12.0	1.097	0.365%	0.100%

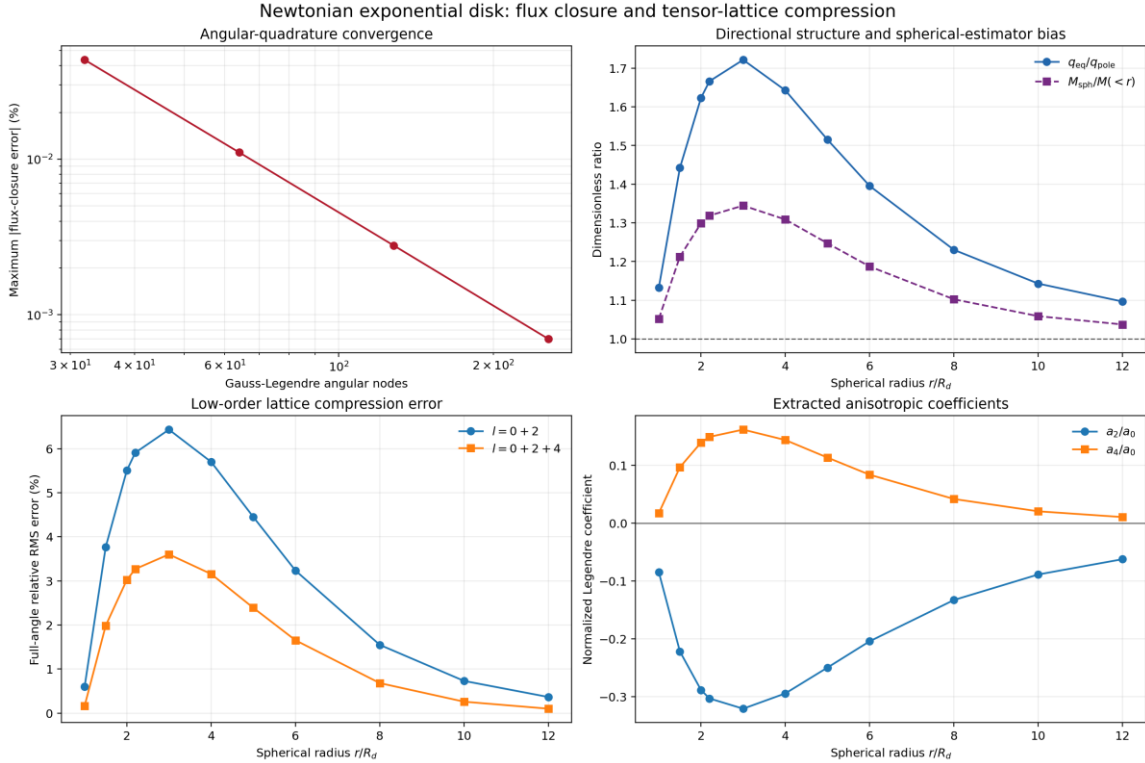


Figure 4. Convergence audit for the exact Newtonian angular flux of an exponential disk and its low-order Legendre/tensor-lattice compression. The monopole records enclosed source and total flux; the higher even modes compress directional structure from the same Newtonian field and are not independent predictions of extra gravity.

The maximum absolute flux-closure error decreases monotonically with angular resolution and reaches 0.000699% at 256 nodes. Tightening the Hankel-integration tolerance changes the normalized coefficients by at most 1.191×10^{-10} . Near $3R_d$, the $l = 0 + 2$ representation has a 6.432% full-angle RMS error; adding $l = 4$ lowers it to 3.602%. By $12R_d$ the corresponding errors fall to 0.365% and 0.100%. A low-order lattice is therefore an auditable compression, especially in the far field, but it should not replace the complete Newtonian disk calculation in the main dynamical region.

Applying the spherical estimator $M_{\text{sph}} = v^2 r / G_0$ to the exact equatorial disk field gives $M_{\text{sph}}/M(<r) = q_{\text{eq}}/a_0$. On the sampled grid this ratio reaches 1.345 at $3R_d$, an apparent overstatement

of 34.5%. The offset is generated entirely by ordinary Newtonian source geometry. It is neither additional mass nor an independently inferred geometrical response, and because a proper SPARC baryonic field calculation already contains disk geometry, it cannot be deducted again from a dark component.

Numerical lesson: Ordinary Newtonian disk geometry already produces a substantial equal-radius equatorial enhancement. Re-fitting it with a tensor does not add missing mass and does not establish a galactic spatial metric with ellipsoidal level sets. A new empirical target would have to be a residual T_R after the complete Newtonian baryonic field has been calculated.

5.13 A neutral residual response for galaxy rotation curves

SPARC baryonic accelerations are computed from the observed distributions of stellar disks, bulges, and gas rather than from a spherical point-mass approximation. Source geometry is therefore already present in $g_{\text{bar},0}(r)$. Multiplying that result by a quantity called J_R would prejudge the residual as geometry and risk counting the disk twice. The empirical stage instead uses a neutral scalar response J_R :

$$g_{\text{pred}}(r) = (\varepsilon_{\text{ref}}/\varepsilon_{g0,S}) g_{\text{bar},0}(r)/J_R(r; \theta), \quad v_{\text{pred}}^2(r) = r g_{\text{pred}}(r)$$

Promotion gate: $J_R \rightarrow J_\Gamma$ only if J_R can be derived from an independent geometrical theory or independently constrained geometry. The evidential order is phenomenological response $\rightarrow$ independent structural derivation $\rightarrow$ possible geometrical interpretation.

5.14 The local-slope identity and the asymptotic test

$$d \ln J_R / d \ln r = d \ln M_{\text{eff}} / d \ln r - 1 - 2 d \ln v_c / d \ln r$$

Here $M_{\text{eff}}(r) = 4\pi\varepsilon_{g0,S} r^2 g_{\text{bar},0}(r)$ is a field-equivalent source measure constructed from the published baryonic acceleration. Inside a non-spherical disk it need not be the literal enclosed mass. The strict $J_R \propto r^{-1}$ asymptote follows only where both $|d \ln M_{\text{eff}} / d \ln r| < \delta_M$ and $|d \ln v_c / d \ln r| < \delta_v$. In that jointly mass-converged and velocity-flat region, the response-area proxy $A_R = 4\pi r^2 J_R$ grows approximately linearly with r . This converts a hidden asymptotic assumption into a pointwise diagnostic.

If J_R is defined directly as $V_{\text{bar}}^2/V_{\text{obs}}^2$ at the same points, the equation becomes an algebraic identity rather than a predictive test. A proper test freezes a model for J_R on training galaxies and then evaluates the slope residual on held-out systems. A slope reconstructed from the same observed curve cannot count as independent confirmation.

5.15 Two BTFR branches: informed calibration and structure-first prediction

$$J_{R,n}(r; r_R, n) = [1 + (r/r_R)^n]^{-1/n}, \quad r_R \propto M_b^\alpha$$

$$V_f^2 = G_0 M_b / r_R, \quad s_{\text{pred}} = 2/(1 - \alpha) \quad \text{for } M_b \propto V_f^{s_{\text{pred}}}$$

In the BTFR-informed calibration branch, an observed slope s is allowed to set $\alpha = 1 - 2/s$. The SPARC analysis of Lelli et al. (2019) reports $s = 3.85 \pm 0.09$ for one commonly used velocity definition, corresponding to $\alpha \approx 0.48$; the idealized $s = 4$ choice gives $\alpha = 1/2$. In this branch the slope is a calibration constraint. Held-out galaxies may test normalization, scatter, transfer across mass, and rotation-curve shape, but the BTFR slope cannot be advertised as newly predicted.

The structure-first branch requires α to follow from a principle or measurement independent of rotation curves and the BTFR. Only then is s_{pred} a prediction. The two branches formalize the paper's maxim that a reparameterization or calibrated restatement is not a prediction.

5.16 Exploratory SPARC calculation and a negative result

The protocol was executed as an exploratory, non-preregistered calculation after the SPARC and radial-acceleration-relation literature had been examined (Lelli et al. 2016; McGaugh, Lelli, and Schombert 2016). SPARC quality classes $Q_{\text{SPARC}} = 1$ or 2 yielded 153 galaxies and 3,166 radial points. Thirty 80/20 splits were performed at the galaxy level, and errors were aggregated with equal galaxy weight in log velocity. The comparison included a baryon-only baseline, a fixed RAR benchmark, $\alpha = 0.48$ and $\alpha = 1/2$ response branches, a freely fitted α , a freely fitted outer exponent, and a disk-scale transition family. These are method diagnostics, not a blind confirmatory test.

Table 5. Exploratory galaxy-level SPARC holdout comparison (equal-galaxy log-velocity RMSE)

Model	Median RMSE (dex)	held-out	Interpretation
Fixed RAR	0.09870		Best benchmark; lower error than the free-mass response in all 30 splits
Free α	0.11919		Median $\alpha \approx 0.624$
$\alpha = 1/2$	0.12129		Idealized BTFR-calibrated branch
$\alpha = 0.48$	0.12181		BTFR-informed branch
$\alpha = 0.48$, free p	0.12187		Extra freedom did not improve prediction
$r_R \propto R_d$	0.14024		Disk-scale transition branch
Baryon only	0.27683		Negative-control baseline

The simple residual families markedly improved on the baryon-only baseline, but none outperformed the fixed RAR benchmark. Across the repeated splits, the fixed RAR had lower error in all 30 comparisons with the free- α and disk-scale families and in 29 of 30 comparisons with each fixed- α family. The free- α median was about 0.624, with an 80% interval of approximately 0.595–0.642. A free outer exponent produced no predictive gain and worsened the local Jacobian condition number from about 7.6 to about 24. The appropriate result is therefore negative but informative: the present ansatz cannot be described as a successful joint prediction of rotation curves, BTFR, and RAR.

With exploratory thresholds $\delta_M = \delta_v = 0.10$, only 78 points in 31 of 153 galaxies satisfied both approximate mass convergence and velocity flatness. They formed 43 contiguous intervals, only ten of which contained at least three points. A principal held-out realization contained 19 eligible points in five galaxies and only two intervals with at least three points. The frozen model's slope residual against the local identity has an RMSE of about 0.403 in the full eligible set and 0.371 in the held-out set. Derivative uncertainties were not propagated, so this is not a formal rejection; it is sufficient to show that an observed 'flat part' should not automatically be treated as $r \rightarrow \infty$.

Within the same BTFR table, the fitted slope $s = 3.7541$ implies $\alpha = 1 - 2/s = 0.467249$. Directly regressing $r_R = G_0 M_b / V_f^2$ gives $\alpha = 0.467249$, differing by only 2.22×10^{-16} . This machine-precision identity exposes the circular-prediction risk directly and motivates the two-branch separation above. The separately frozen $\alpha = 0.4805$ branch remains explicitly labelled as informed by the published $s \approx 3.85$ relation. A future test should be called a prospective prespecified or proposed locked analysis protocol—not preregistered unless a version is publicly registered before execution—and should ultimately use a galaxy sample that did not participate in model construction.

5.17 Conditional mass-bias hypothesis and non-unique inversion

$$M_{\text{dyn}}^N(r) = r^2 g_{\text{obs}} / G_0 = 4\pi \varepsilon_{g0,s} r^2 g_{\text{obs}}$$

$$M_{\text{exc}}^N = M_{\text{dyn}}^N - M_{\text{bar}}$$

The superscript N marks a Newtonian-equivalent spherical inversion. If $g_{\text{obs}} = q_R g_{\text{bar}}$ and the same simple spherical source measure is used, then $M_{\text{exc}}^N = (q_R - 1)M_{\text{bar}}$. For a disk, the safer comparison is field-equivalent: $M_{\text{eff}} = 4\pi \varepsilon_{g0,s} r^2 g_{\text{bar}}$, because this quantity need not equal a literal enclosed baryonic mass. A dynamical excess inferred under a chosen field equation is real as an inverse-model result, but it does not identify its ontology.

$$H_{\text{mass}}: g_{\text{obs}} = g_{\text{bar}} + g_{\text{DM}}; \quad H_{\text{response}}: g_{\text{obs}} = q_R g_{\text{bar}}$$

These alternatives can be arranged to agree on equatorial circular speeds while disagreeing in polar or vertical acceleration, weak lensing, stream precession, time evolution, and conservation residuals. The scientific target is consequently not the universal proposition 'dark matter does not exist.' Dark-matter model families are empirically testable, and a universal negative would outrun the evidence. The tractable claim is that dynamical mass discrepancy does not uniquely entail extra matter, and that a residual field-response model can be compared with extra-mass models by observations on which their predictions diverge.

A spherical inversion of the exact Newtonian exponential-disk field overstates the enclosed baryonic mass by as much as 34.5% at $3R_d$ on the audited radial grid, but the bias decreases rapidly outward. Inserted into the outer SPARC discrepancy budget, the median contribution is only about 3.73% for 111

galaxies with measurements at or beyond $4R_d$, about 4.61% overall in that subset, and about 2.77% beyond $8R_d$. Ordinary disk source geometry can therefore create a finite inference bias but cannot account for the full outer discrepancy. Moreover, SPARC's published V_{bar} already includes disk geometry, so this correction cannot be subtracted a second time.

If an independently established residual equatorial focusing q_F remained after the complete source model, a simple estimate of its share of the apparent discrepancy would be $f_{\text{geo}} = (q_F - 1)J_b / (1 - J_b)$, where $J_b = g_{\text{bar}} / g_{\text{obs}}$. At $J_b = 0.22$, $q_F = 1.1, 1.2, 1.5$, and 2 would contribute approximately 2.8%, 5.6%, 14.1%, and 28.2%, respectively; explaining the entire discrepancy would require $q_F \approx 1/J_b \approx 4.5$ and a correspondingly strong off-plane compensation. These numbers define a conditional bias hypothesis, not an inference from present rotation curves.

Ontological restraint: The defensible statement is 'a residual field response could contribute to a Newtonian-equivalent excess under specified conditions,' not 'dark matter has been disproved' and not 'the inferred dark-matter mass is known to be overestimated.'

5.18 A complementary conservation–sequence rank experiment

A separate synthetic experiment tests whether conservation and lawful history can turn a static residual into a more discriminating research program. Nine hundred synthetic galaxies were generated at five epochs, $z = 2, 1.5, 1, 0.5$, and 0 , with 300 noise realizations. Structural flattening q_{seq} evolved from a median of 0.460 to 0.236. The angular response was generated as $Q_{\text{seq}}(\theta, t) = B[1 - \kappa(1 - q_{\text{seq},t})P_2(\cos \theta)]$, with $B = 1$ enforcing mean-flux conservation and $\kappa = 0.36$ controlling anisotropic coupling. The first four epochs trained the historical relation; the final epoch was held out.

Table 6. Synthetic conservation–sequence holdout and negative-control results

Audit quantity	Isotropic model	Conservation–sequence tensor model	Interpretation
Equatorial holdout RMSE	0.05560	0.05006	Small gain in the channel used by ordinary rotation curves
Joint equatorial + polar RMSE	0.23196	0.05006	Large gain when a nonparallel channel is present
Recovered κ	—	0.35996 [0.35892, 0.36119]	Recovers the imposed $\kappa = 0.36$ under the true generator
Isotropic null, recovered κ	0	−0.00005 [−0.00125, 0.00137]	No false anisotropy under $B = 1.18, \kappa = 0$
Final-epoch κ shock	—	1.992 noise σ RMS	Detectable sequence violation

Table 7. Rank audit for the synthetic conservation–sequence experiment

Design	Jacobian size	Rank	Nullity / condition
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Design	Jacobian size	Rank	Nullity / condition
Independent epochs, equatorial means	4×8	4	Nullity 4
Independent epochs, joint means	8×8	8	Nullity 0; condition ≈ 1.62
Locked sequence, equatorial means	4×2	2	Nullity 0; condition ≈ 22.79
Locked sequence, joint means	8×2	2	Nullity 0; condition ≈ 1.59

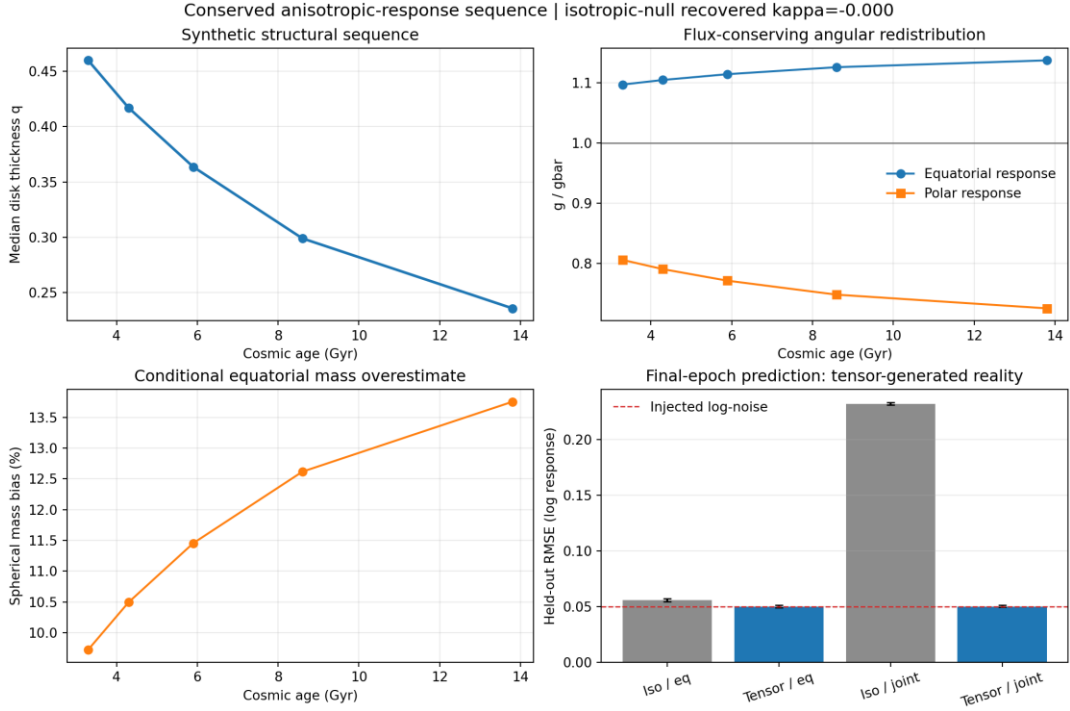


Figure 5. Five-epoch synthetic conservation-sequence rank experiment. The data are generated by the stated toy law; the figure tests the analysis design and its negative controls, not the geometry of real galaxies.

The central result is not that the toy tensor is true. Equatorial observations alone show only a modest held-out advantage and remain poorly conditioned after sequence locking. Joint equatorial and polar observations recover the imposed anisotropy stably, reject the isotropic null without producing false anisotropy, and expose a final-epoch shock as a sequence residual. Conservation reduces admissible freedom, sequence linkage replaces independent epoch parameters with a shared law, and a nonparallel channel supplies the empirical direction. The three operations are complementary but logically distinct.

Because the data were generated by the assumed law, recovery of κ validates code, identifiability logic, negative controls, and a prospective falsification design. It does not establish that real galaxies obey the law. The new methodological contribution is a constrained generative comparison: competing hypotheses that are observationally equivalent in one channel at one epoch should be propagated through explicit state transitions and observation operators until they diverge in held-out spatial or temporal channels.

5.19 Existing observations and the status of ellipsoidal equipotentials and spatial geometry

There is no direct observation that a galactic spatial metric has globally ellipsoidal level sets. Three evidential levels must be separated. Imaging directly shows that stellar and gas sources are often flattened or triaxial. Stellar streams, polar structures, vertical dynamics, and weak lensing can infer non-spherical mass distributions or gravitational potentials under explicit models. Within the declared constant-coefficient toy operator, ellipsoidal equipotentials follow analytically; within the finite-time generator, nonspherical response is conditionally co-generated by source evolution and closure. Neither result is a direct measurement of a three-dimensional spatial metric, and isodensity, equipotential, equal- $|g|$, and metric-curvature structures remain non-equivalent.

As examples, a Gaia DR3 stream-based Milky Way model reported a dark-halo density flattening $q_{m,h} = 0.75 \pm 0.03$ under its adopted double-power-law mass model (Ibata et al. 2024). A stacked weak-lensing analysis of 10,428 SDSS clusters inferred a projected halo axis ratio of approximately 0.56 ± 0.09 (statistical) ± 0.03 (systematic) under its alignment and projection assumptions (Shin et al. 2018). These results support non-spherical mass modeling. They do not identify a tensor-lattice parameter β , establish a galactic metric with ellipsoidal level sets, or show that a residual response replaces matter. The shape of a gas nebula is still less diagnostic because magnetic fields, pressure, radiation, shocks, and turbulence can dominate its morphology.

More strongly pinched, annular, or topology-changing equipotential branches would require additional source, rotation, or closure structure and are not generated or claimed by the present numerical cases.

Evidence level: Current evidence supports non-spherical sources and model-dependent non-spherical gravitational potentials. Ellipsoidal equipotentials may arise as conditional consequences of specified anisotropic operators or self-consistent source–field closures. A galactic spatial metric with corresponding ellipsoidal level sets remains a distinct hypothesis subject to an independent geometrical promotion gate.

5.20 Falsifiability and the strong-field boundary

The galaxy branch fails if it cannot predict held-out curves, persistently violates the local-slope condition in legitimately converged regions, needs an unrestricted response for every galaxy, or leaves the response sensitivity inside the nuisance column space. The tensor branch additionally predicts cross-direction relations: equatorial and polar acceleration, lensing quadrupoles, stream precession, or vertical dynamics must agree with one common response. A tensor that merely re-fits the Newtonian disk has no independent empirical content.

Black holes are not a validation case in this manuscript. The $\varepsilon_{g0,s}-T_R$ equation is defined only as a static or quasi-static weak-field representation. It cannot replace Einstein's equation or compress Kerr

spacetime into a Euclidean ellipsoid. A strong-field extension would require a covariant action or metric-derived operator, the Bianchi identities and energy–momentum conservation, and consistent treatment of spin, null propagation, polarization, and radiation. Strong-field systems therefore delimit the present proposal rather than confirm it.

5.21 Chapter lesson

The gravitational example illustrates the entire method. Separating $\varepsilon_{g0,s}$ from Γ makes explanatory provenance visible; the constraint equation and flux law prevent the separated terms from becoming free inventions; J_R and T_R avoid assuming that a residual is geometrical; BTFR calibration is kept distinct from prediction; the Newtonian disk audit blocks double counting; likelihood rank is reported before prior-conditioned stability; and the conservation–sequence simulation turns a verbal historical story into a held-out generative test. The result is an alternative explanatory program, not an ontological verdict.

6. Free Fields and Constrained Transport

6.1 Two questions about the same charge

The statements 'an isolated point charge has a spherically symmetric electrostatic field' and 'electric current flows along a wire' may seem to attribute incompatible natural forms to charge. The appearance of conflict results from conflating two different questions. The first concerns the spatial configuration of a field generated by an idealized source in a symmetric environment. The second concerns the macroscopic transport of many carriers through a material channel under an applied potential difference.

6.2 Field dimension and transport dimension

$$d_{\text{field}} \neq d_{\text{transport}}$$

A thin wire is embedded in three-dimensional space. Its macroscopic current may be modeled as one-dimensional, but its electric and magnetic fields remain three-dimensional. The magnetic field around an ideal straight current, for example, forms circles around the wire and exhibits axial symmetry. The dimensionality of the carrier flux is therefore not the dimensionality of the surrounding field.

6.3 How a conductor produces quasi-one-dimensional current

$$\mathbf{J} = \sigma \mathbf{E}$$

In an Ohmic medium, the local current density $\mathbf{J}$ is approximately proportional to the electric field $\mathbf{E}$. The geometry of the conductor, the distribution of surface charges, the material's conductivity, and the external circuit jointly establish an electric field with a dominant tangential component. Persistent

transverse transport is suppressed by the material boundary, whereas the longitudinal component accumulates into a macroscopic net current.

$$\mathbf{J} = nq\mathbf{v}_d$$

Microscopically, charge carriers do not travel along smooth straight lines. They undergo thermal motion and repeated scattering from lattice vibrations, impurities, and defects. The phrase 'current flows straight' refers to the direction of the averaged drift velocity, not to each microscopic trajectory. Random transverse components largely cancel at the macroscopic scale, and the boundary prevents a sustained net flux across the conductor surface.

6.4 Boundary as a producer of effective form

The conductor does not compress the entire electromagnetic field into a line. It constrains the effective degrees of freedom of material transport. A straight wire supports an approximately straight macroscopic current, a curved wire supports current along its curvature, and a ring supports a closed circulation. The geometry of the channel is thus not a decorative container added to a pre-existing one-dimensional current; it helps constitute the current's macroscopic form.

**Spherical symmetry characterizes the free field of an ideal point source;
linearity characterizes transport constrained by a slender material boundary.**

This case supports a broader philosophical point. Boundaries can be generative. They do not merely obstruct an otherwise natural behavior; by suppressing some degrees of freedom and stabilizing others, they make new effective regimes possible. Waveguides, thin films, cavities, and one-dimensional channels all exhibit this structure.

7. Cosmology as a Natural Observational Sequence

7.1 Why cosmology is a historical physics problem

Cosmology provides a stronger test of the historical component of the framework than heat diffusion because the relevant process is not repeatable as a laboratory sequence. We do not observe the universe being rerun from an early hot state to the present. Instead, contemporary cosmology reconstructs an ordered history from physically different traces associated with different epochs. Light-element abundances, relic radiation, large-scale structure, high-redshift objects, gravitational lensing, supernova distances, and other observables constrain different parts of one evolving system. The explanatory problem is therefore not only whether each observation can be fitted locally, but whether the inferred states can belong to one physically admissible history (Peebles 1993; Weinberg 2008; Planck Collaboration 2020).

The cosmological example gives a precise meaning to History in the title of this paper. History is not an additional force acting alongside geometry and matter. It is state inheritance across a constrained sequence: what a later cosmic epoch can become depends on what earlier evolution has already made physically available.

7.2 Natural observational sequence

For the present analysis, a natural observational sequence (NOS) is an ordered relation among reconstructed physical states, admissible transitions, and epoch-specific observables in a process that cannot be experimentally rerun in its entirety. Let $X_0, X_1, \dots, X_n$ denote reconstructed cosmic states and $O_0, O_1, \dots, O_n$ the observational traces associated with them. The minimal structure is

$$\begin{aligned} X_0 &\rightarrow X_1 \rightarrow X_2 \rightarrow \dots \rightarrow X_n, \\ X_i &\rightarrow O_i. \end{aligned}$$

The first relation represents physical succession under a specified cosmological model; the second represents the production of observable traces by a state. This notation is not a claim that cosmology is literally a hidden Markov model. It is an audit representation that separates the state-transition question from the observation-generation question. A theory may fit one observation while failing to provide a viable bridge between the states required by two others. Such a failure is a sequence inconsistency, not merely a low score on an isolated datum.

7.3 Historical transfer and state inheritance across cosmic epochs

At a cosmological epoch, the sufficient state may include the scale factor, expansion rate, source content, perturbations, and ionization variables required by the model. Later states are constrained by the earlier state and transition law. This historical transfer may involve propagation, damping, growth, mode conversion, conservation, or changing observational access; it should not be identified automatically with a non-Markovian kernel. A late-time modification must remain compatible with the earlier stages through which the model claims the universe passed.

7.4 Constraints and admissible historical pathways

The generic dynamic skeleton can now be read cosmologically. Let $\mathcal{A}_t$ denote the states and transitions admitted at epoch t by the metric structure, matter content, conservation relations, field equations, and any declared model constraints:

$$\begin{aligned} (\Gamma_t, S_t, B_t) &\rightarrow \mathcal{A}_t, \\ (X_t, I_t; \mathcal{A}_t) &\rightarrow X_{t+1} \rightarrow O_{t+1}. \end{aligned}$$

Here B need not mean a literal material wall. In cosmology it may represent an admissibility condition, matching condition, topology, approximation regime, or other restriction that limits the

model's allowed transitions. The notation therefore generalizes the boundary idea from laboratory systems to a constraint on the space of viable cosmic histories.

A viable cosmological explanation must fit not only individual observations but the admissible historical sequence connecting them.

7.5 Structure-explicit reading of the Friedmann equation

The cosmological extension should not force the radial factor J_r into a context where it is not derived. In a standard Friedmann-Lemaître-Robertson-Walker model, taking $c = 1$ for compactness, the first Friedmann equation may be written

$$\begin{aligned} H_{\text{Hub}}^2 &= (8\pi G_0/3)\rho - k/a^2 + \Lambda/3 \\ &= 2\rho/(3\varepsilon_{g0}) - k/a^2 + \Lambda/3, \quad \varepsilon_{g0} = 1/(4\pi G_0). \end{aligned}$$

This algebraic rewrite does not add a new degree of freedom. Its value is diagnostic: the observable expansion history is related simultaneously to source content, geometry, and gravitational response. Unlike the reduced radial example, however, the Friedmann equation already displays several of these contributions explicitly. The cosmological task is therefore not primarily to replace $4\pi r^2$ by a new area factor, but to preserve the provenance of quantities that are otherwise compressed into an inferred parameter or a model-dependent residual.

For this reason, cosmology represents a higher level of explanatory decompression: coefficient decomposition in the Newtonian example becomes model-structure decomposition in an evolving spacetime.

7.6 Geometry-first null hypothesis across cosmic history

The geometry-first ordering can be generalized without claiming that G varies. Let $H_{r,\text{cos}}$ denote the null hypothesis that G_0 and ε_{g0} remain constant while the geometry, source content, and inherited cosmic state evolve according to the chosen cosmological theory. For an observable $\mathcal{O}$,

$$\begin{aligned} H_{r,\text{cos}}: G &= G_0, \quad \varepsilon_g = \varepsilon_{g0}, \\ \mathcal{O}_{\text{pred}} &= F(I_t, S_t, X_t, B_t; G_0). \end{aligned}$$

A stronger coupling hypothesis becomes relevant only when a reproducible residual survives an independently specified geometry, source model, historical pathway, observational calibration, and model-systematic audit. A proposal such as $G = G(t)$ must therefore do more than improve one late-time scalar. It must generate a coherent sequence through the earlier epochs that are independently constrained by other observations.

7.7 Sequence-linked rank restoration and conditional ε_{g0} decoding

Repeated epochs do not identify a shared normalization if each epoch receives an independent structural parameter. A lawful transition model can replace those nuisance parameters with a smaller shared set, but data rank is restored only when the joint likelihood contributes nonparallel sensitivities. BBN, CMB, BAO, supernovae, and lensing can do so under declared cosmological forward models because they couple gravity to different microphysics and observation operators.

$$\ln \varepsilon_g(z) = \ln \varepsilon_{g0} + q_{\cos}(z; \varphi), \quad q_{\cos}(0; \varphi) = 0$$

The anchor prevents a trivial normalization trade but is only an identifiability device, not a dynamical field theory. A spacetime-dependent response requires additional dynamics and compatible conservation laws. Sequence linkage reduces admissible freedom; the data Jacobian determines whether the shared response is actually identified.

7.8 Hubble-constant tension as divergence between inference pathways

The well-known disagreement between a CMB-based inference of the present Hubble constant under a specified cosmological model and late-universe distance-ladder determinations provides an instructive example of structural divergence (Planck Collaboration 2020; Riess et al. 2022). The useful point here is methodological rather than numerical. The CMB route does not directly observe today's Hubble constant at recombination; it infers a present parameter by propagating information through a cosmological model. A distance ladder reaches the same target through a different observational chain.

The discrepancy can therefore be represented schematically as two natural inference sequences terminating in estimates of one present-state quantity:

$$\begin{aligned} O_{\text{CMB}} &\rightarrow X_{\text{rec}} \rightarrow \text{model transition history} \rightarrow H_{0,\text{inferred}}, \\ O_{\text{local}} &\rightarrow \text{calibrated distance sequence} \rightarrow H_{0,\text{local}}. \end{aligned}$$

A difference between the endpoints is diagnostically important, but it does not identify its own cause. Geometry, source content, calibration, observational selection, transition assumptions, or a genuinely different gravitational response can occupy different causal roles. Structural analysis asks where the sequences first become mutually inconsistent before granting the final scalar discrepancy fundamental causal authority.

7.9 Pseudo-longitudinal galaxy populations

Cosmology displays populations at different cosmic times and therefore provides a natural space for testing historical predictions. It does not, however, provide a continuous film of the same galaxy. Progenitor bias, mergers, rank reordering, mass completeness, surface-brightness selection, beam

smearing, environment, and mass-to-light modeling can make a sequence of cross-sections look like evolution. The appropriate object is a population transition kernel rather than a literal individual trajectory:

$$p(X_{t+\Delta t}) = \int K_{\theta}(X_{t+\Delta t} | X_t) p(X_t) dX_t$$

A strong test should proceed forward in cosmic time: estimate the earlier distribution and a frozen transition rule from higher-redshift bins, then predict a lower-redshift population not used to choose the ansatz. Fixed-mass, constant-number-density, mass-rank, and simulation-lineage matching are competing progenitor models and should be declared in advance rather than selected after seeing the lowest residual.

7.10 A synthetic pseudo-longitudinal audit

A five-epoch synthetic population experiment at $z = 2, 1.5, 1, 0.5$, and 0 used 2,400 latent lineages, redshift-dependent selection, and a state containing baryonic mass, disk scale, thickness, gas fraction, isotropic excess, and residual quadrupole. Twenty-four per cent of selected objects supplied a polar or vertical channel; the final epoch was held out. Separate generators represented conserved anisotropy, isotropic dark matter, and time-varying isotropic excess. The audit used 120 repetitions and varied nonparallel-channel coverage.

Table 8. Final-epoch prediction in 120 synthetic pseudo-longitudinal populations

Final-epoch strategy	Median RMSE	16%–84% interval
Independently locked sequence	0.00871	0.00606–0.01210
Static structural model	0.01240	0.00705–0.02052
Linear time extrapolation	0.01496	0.01089–0.01952
Persistence baseline	0.01927	0.01478–0.02417
Freely fitted in-sample sequence	0.02011	0.01084–0.02957

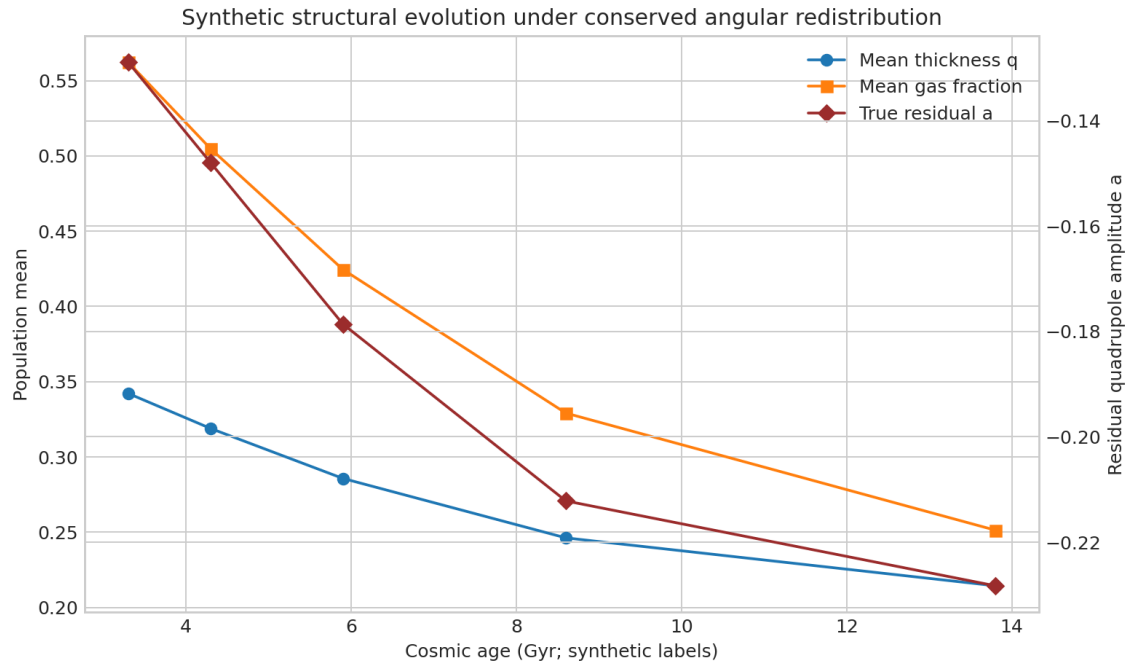


Figure 6. Structural evolution in the synthetic galaxy populations. The residual quadrupole is a generated variable, not a measurement of real galaxies.

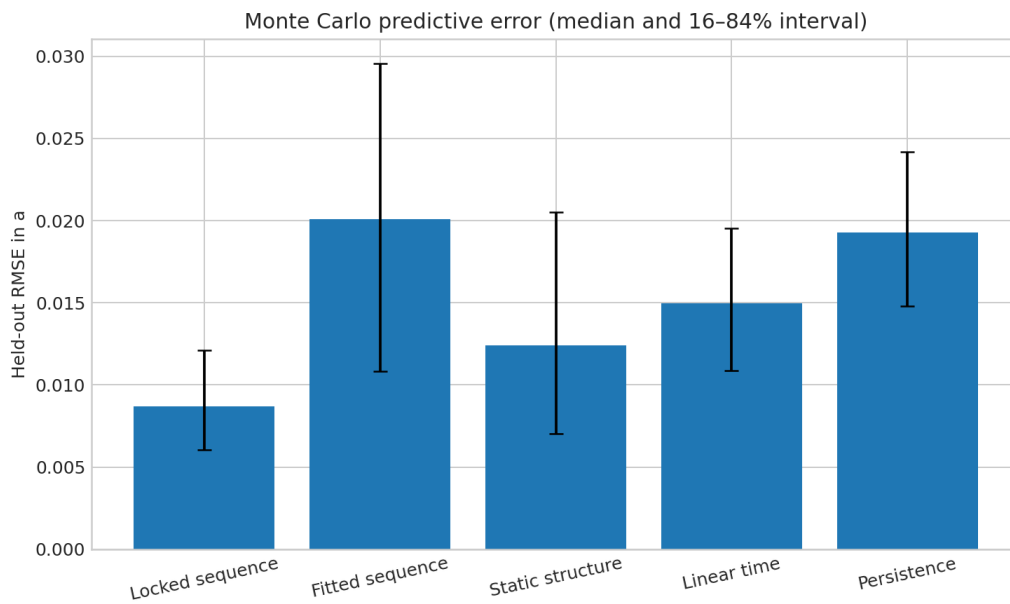


Figure 7. Monte Carlo errors for five final-epoch prediction strategies. The locked sequence uses coefficients fixed independently of the generated fitting sample; the free sequence re-estimates them from the training epochs.

The independently locked sequence reduced median error by about 30% relative to the static model and 55% relative to persistence; the freely fitted sequence did not win reliably. Its full-rank design remained poorly conditioned (about 38) because structural covariates were collinear. Increasing

nonparallel-channel coverage from 12% to 50% reduced its median RMSE from about 0.0273 to 0.0100. Full algebraic rank is therefore not practical identifiability, and redshift ordering is not historical linkage.

7.11 What cosmology adds to the framework

Cosmology adds historical-dynamical decompression to coefficient and geometrical decompression. An inference path must specify not only which roles produce an observable at one epoch, but also which transition law and observation operator connect populations and traces across epochs. The CMB, BBN, BAO, supernovae, lensing, and galaxy dynamics can restore rank when they contribute nonparallel physical sensitivities; lawful sequence linkage can further reduce epoch-specific freedom. Neither operation licenses a causal or ontological conclusion until selection effects, nuisance directions, and held-out predictions are audited.

8. Inertia, Recurrence, and the Inheritance of Motion State

8.1 The structural content of Newton's first law

Newton's first law states that a body remains at rest or continues in uniform rectilinear motion unless compelled to change that state by impressed forces. In modern notation, for a constant-mass particle in an inertial frame,

$$\mathbf{F}_{\text{net}} = 0 \quad \Rightarrow \quad \frac{d\mathbf{v}}{dt} = 0$$

$$\mathbf{x}(t) = \mathbf{x}_0 + \mathbf{v}_0 t$$

The law does more than describe a special case of zero acceleration. It helps characterize inertial frames and establishes the baseline against which acceleration is attributed to forces rather than to the coordinate system. In flat Newtonian spacetime, the spatial form of free motion is rectilinear and the velocity is constant with respect to inertial time (Newton 1999; Earman 1986).

8.2 Geometry specifies continuation, not the particular velocity

Flat inertial geometry permits indefinitely many uniform motions. It does not determine whether a given body moves east or west, slowly or rapidly. The concrete motion requires an initial position and velocity. Geometry specifies the form in which an already given inertial state continues; it does not by itself select the state that is continued.

Geometry governs the form of free continuation; initial state governs the concrete content of that continuation.

8.3 Momentum as the integral result of interaction

$$\frac{d\mathbf{p}}{dt} = \mathbf{F}$$

$$\mathbf{p}(t) = \mathbf{p}(t_0) + \int_{t_0}^t \mathbf{F}(\tau) d\tau$$

For an actual body, its present momentum commonly results from an initial momentum together with accumulated impulse. A rocket continues after engine shutdown because the engine has already altered the rocket's momentum. A ball moves after a collision because the collision has changed its state. Once the force ceases, the force no longer exists as a present cause, but its effect remains embodied in momentum.

This motivates a more precise claim about state inheritance. Classical mechanics allows initial velocity to be specified at the boundary of a model without deriving it from an earlier force within that model, so no infinite explanatory regress is required. The philosophically careful statement is that a concrete inertial state is specified by initial data and is often, in real systems, the retained result of previous interactions. History matters because its dynamically relevant consequences are instantiated in the present state, not because a past force continues to act after it has ceased.

8.4 No present force does not imply no causal history

The absence of a present net force answers the question of why the velocity is not currently changing. It does not answer why that velocity has its particular magnitude and direction. These are distinct explanatory questions. Newton's first law addresses persistence; inherited state specifies the concrete motion at the chosen modeling boundary, and earlier interactions may explain how that state was formed.

This distinction also prevents a mistaken opposition between law and history. History is not an exception to law, and law is not a replacement for history. Earlier interactions occurred lawfully and produced states that later laws continue to govern. The present inherited state is the interface through which historical contingency enters a law-governed evolution; in a genuinely sequential problem, the admissibility of the path to that state is a further explanatory requirement.

8.5 First and second laws as different explanatory levels

The second law specifies how force changes momentum. The first law specifies the inertial baseline and the persistence of motion when no net force acts. Although in elementary algebra one can set $F = 0$ in $F = ma$, the conceptual role of the first law is not exhausted by this substitution. It identifies inertial structure and distinguishes genuine acceleration from artifacts of non-inertial description. Within the present framework, the first law belongs primarily to the geometry and state-preservation layers, while the second law belongs to the current-interaction layer.

8.6 Newtonian motion as a sequence-compatibility example

The sequence language can be made concrete without treating discrete recurrence as fundamental physics. Sample uniform inertial motion at equal time intervals Δt . From $x(t) = x_0 + vt$ one obtains

$$x_n = x_0 + nv\Delta t, \quad x_{n+1} - x_n = v\Delta t.$$

The sampled positions therefore form an arithmetic progression. This is not a new reading of the first law; it is a transparent representation of the fact that the same inertial state is compatible with every successive time slice when no net force changes it. The future continuation preserves the state relation already instantiated in the present.

For constant acceleration, the positional sequence is not a geometric progression. Equal-time sampling yields a constant second difference:

$$x_{n+1} - 2x_n + x_{n-1} = a(\Delta t)^2.$$

The second law therefore illustrates a higher-order compatibility relation in the position-only representation. Yet the apparent “memory order” is representation-dependent: the same second-order equation can be written as a first-order system for the phase-space state (x, v) . The physically important point is not the number of previous terms but whether the chosen state representation contains enough information for lawful continuation.

Wave propagation supplies a still richer illustration. A standard centered finite-difference form of the one-dimensional wave equation can be written schematically as

$$u_j^{n+1} = 2u_j^n - u_j^{n-1} + (c\Delta t/\Delta x)^2(u_{j+1}^n - 2u_j^n + u_{j-1}^n).$$

The future time layer depends on the present layer, a previous temporal layer, and neighboring spatial states. This is structurally reminiscent of recurrence sequences such as the Fibonacci rule in the limited sense that later terms are constrained by earlier terms; the physical coefficients and couplings, however, are supplied by the governing law rather than by numerical analogy. Recurrence is therefore a useful visualization of sequence compatibility, not a claim that physical laws are Fibonacci-like at a fundamental level.

A physical law can be read as a generator of admissible continuations once a sufficient state and its constraints have been specified.

9. Geometry and Inherited State in Relativistic Free Motion

9.1 Geometry-relative free continuation

$$d^2x^\mu/d\tau^2 + \Gamma_{\alpha\beta}^\mu(dx^\alpha/d\tau)(dx^\beta/d\tau) = 0$$

Here $\Gamma_{\alpha\beta}^{\mu}$ denotes the Christoffel symbols of the declared metric, not the general geometry-role label Γ and not the Euler gamma function of §4.2.

A relativistic free body follows a geodesic of the specified spacetime geometry. In Euclidean space the corresponding free path is a straight line; on a sphere the intrinsic geodesics are great circles. These are instances of geometry-relative extremal continuation, not evidence that one visible shape is universally fundamental. Constant-curvature spaces are especially symmetric, but generic spacetimes need not have constant curvature.

9.2 Initial state and mutual coupling

A metric does not select one worldline without an initial event and four-velocity. Geometry determines the admissible continuation rule; the inherited state selects a particular geodesic. In general relativity the relation is also reciprocal: stress–energy helps determine the metric, while the metric constrains matter and light propagation. The role map must therefore permit feedback and joint determination. Its purpose is to preserve explanatory provenance, not impose a one-way causal order on Einstein’s theory.

10. The Structural Determination Framework: Roles, Coupling, and Causal Audit

10.1 The schema and its intended level

$$R = \Phi_{\text{map}}(\Gamma, S, H, B, I).$$

The function symbol Φ_{map} is not an unspecified universal algorithm that competes with physical laws. It marks a relation of explanatory dependence. In a concrete case the relation is instantiated by established equations, constitutive assumptions, a domain, source data, state variables, and boundary conditions. The schema asks which part of that model accounts for which feature of the realized form.

Its variables should therefore be read as roles. The same physical quantity can change role across models, and several roles may be physically coupled. The framework is successful only when that separation clarifies provenance rather than multiplying labels.

10.2 A role-based causal vocabulary

Geometry Γ usually acts as a structural condition, propagation operator, or mediator. Source S acts as a driver or parent variable specifying what is distributed or coupled. History H denotes the explanatory role of inherited state and path dependence; in dynamic form this is represented by X_t rather than by a separate historical substance. Boundary or medium B acts as a constraint, moderator, or admissibility

condition. Current interaction I acts as a transition driver. Realized form R is the downstream configuration to be explained. These labels are model-relative rather than ontological declarations.

These labels are not exclusive. In general relativity stress-energy contributes to geometry, so the chain $S \rightarrow \Gamma \rightarrow R$ can be more informative than treating S and Γ as unrelated inputs. In a conductor, boundary charges alter the electric field that drives transport. In a nonlinear material, inherited state changes constitutive response. Causal coupling is therefore expected; the analytical separation is an audit device, not a doctrine of ontological independence.

10.3 Dynamic admissibility and state transition

$$H_t = R(H_{<t}), \quad X_t = \Psi(Z_t, H_t)$$

$$(\Gamma_t, S_t, B_t; \mathcal{C}_t) \rightarrow \mathcal{A}_t, \quad X_{t+1} = T_t(X_t, I_t; \mathcal{A}_t)$$

The sufficient state may already encode historical consequences; explicit finite memory, $H_t = \Sigma_r K_r Z_{t-r}$, is added only when required by the reduced model. Any finite-order recurrence can be rewritten as first-order evolution on an augmented state. This changes representation, not the history that produced the retained variables. Geometry, source, boundary, and governing constraints determine admissibility; inherited state and current interaction select a continuation. Feedback may update these role variables at the next time.

10.4 From role decomposition to constrained generative comparison

Role identification $\rightarrow$ analytical decomposition $\rightarrow$ governing constraints $\rightarrow$ candidate generators $\rightarrow$ observation operators $\rightarrow$ likelihood-rank audit $\rightarrow$ held-out divergence $\rightarrow$ causal audit $\rightarrow$ controlled recompression

The original static role map can be embedded in a time-indexed generative and observational form:

$$\mathcal{A}_t = \mathcal{A}(\Gamma_t, S_t, B_t; \mathcal{C}_t)$$

$$\mathcal{F}_t \in \mathcal{A}_t$$

$$X_{t+1} = \mathcal{F}_t(X_t, I_t; \theta) + \xi_t$$

$$R_{t+1} = \Psi(X_{t+1})$$

$$O_{t+1,c} = \mathcal{M}_c(R_{t+1}, \text{selection}_{t+1}) + \eta_{t+1,c}$$

Here S_t denotes the source/scalar sector and includes $\varepsilon_{g0,S}$ where applicable, while $\mathcal{C}_t$ collects the governing conservation and regularity constraints. Retained historical information H_t is encoded in the dynamically sufficient state X_t , rather than appended as an independent instantaneous interaction. The condition $\mathcal{F}_t \in \mathcal{A}_t$ means that the candidate transition rule must belong to the space permitted by the declared geometry, source structure, boundary conditions, and governing constraints. The map Ψ

produces the realized physical configuration, whereas $\mathcal{M}_c$ is the channel-specific observation operator, including selection and measurement effects.

Sequence compatibility restricts the admissible transitions and histories, but it does not by itself supply empirical rank. Competing hypotheses may be observationally equivalent in equatorial velocities at a single epoch, $O_{\text{eq}}(H_{\text{mass}}) \approx O_{\text{eq}}(H_{\text{response}})$, while diverging across the joint observational set $\{O_{\text{eq}}, O_{\text{pole}}, O_{\text{lens}}, O_{t+1}\}$. The method therefore seeks combinations of observational channels and epochs for which the relevant target-sensitivity column leaves the nuisance column space, $j_0 \notin \text{Col}(J_\eta)$, so that the role-separated, decompressed explanations cease to be mere reparameterizations of one another.

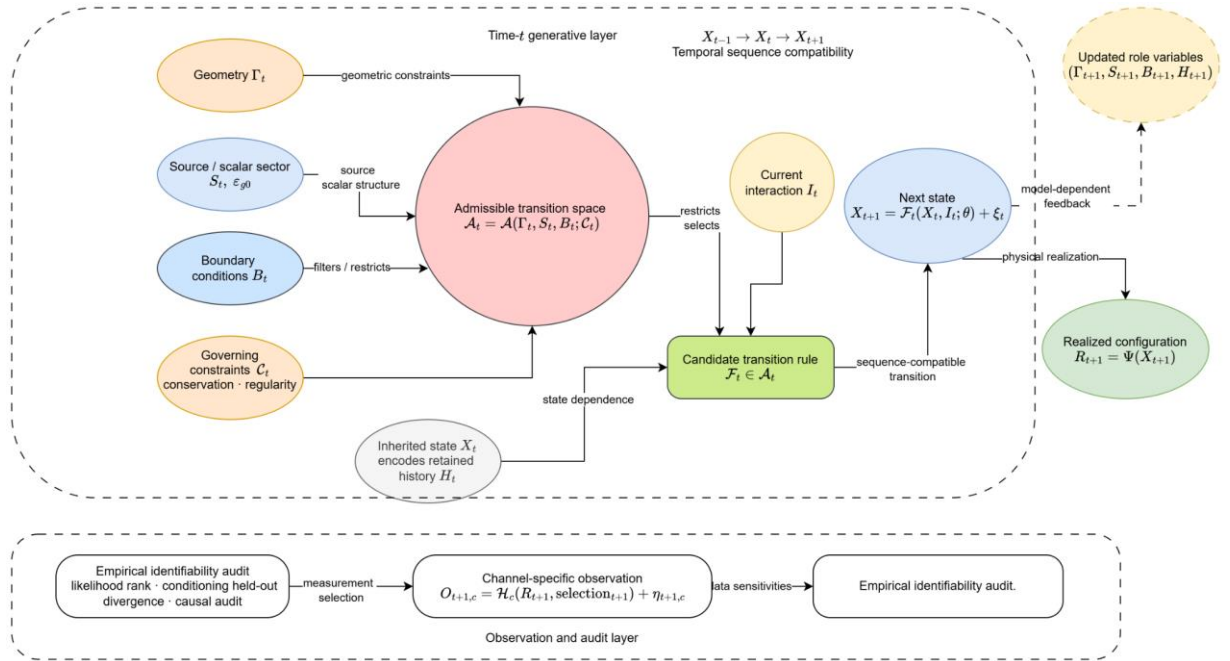


Figure 8. Time-indexed generative and observational architecture. Geometry, the source/scalar sector, boundary conditions, governing constraints, inherited state, and current interaction jointly restrict the candidate transition rule and the next realized configuration. Channel-specific observations then supply the likelihood sensitivities used in the empirical identifiability audit. Sequence compatibility restricts admissible histories but does not by itself supply empirical rank.

Conservation and sequence equations reduce the admissible dimension of a model. They do not, by themselves, create data information. Empirical rank comes from the likelihood sensitivities of actual channels; practical identifiability further depends on noise and conditioning. A model can therefore be dynamically admissible but statistically non-identifying, structurally full rank but practically unstable, or predictively successful without being geometrically or ontologically established.

10.5 Causal authority, proxies, and recompressed coefficients

Sensitivity or correlation does not grant causal authority. A fitted quantity may be a driver, mediator, moderator, proxy, confounded indicator, or downstream recompression. Boundaries may act by restricting the reachable set rather than adding a force. Likewise, G_{eff} can summarize a geometry–response pathway

without identifying its upstream cause. Only independent source calibration, geometrical information, likelihood-rank analysis, and divergent predictions can promote a recompressed coefficient from convenient description to supported explanation.

10.6 Multi-channel rank restoration and sequence-linked rank restoration

Multi-channel rank restoration is the general concept. Same-epoch lensing and dynamics, an independently measured source mass, or equatorial and polar accelerations can supply nonparallel sensitivity directions without any temporal ordering. In a partition $J = (j_0, J_\eta)$, the target direction is likelihood-identifiable only when j_0 is not in $\text{Col}(J_\eta)$, subject to the declared forward model. Priors are reported separately.

Sequence-linked rank restoration is a special mechanism for historical systems. A lawful chain $X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_n$ can replace independent epoch parameters with a smaller shared transition parameter set. This reduces model freedom, but a sequence of repeated observations with parallel sensitivities may remain uninformative about the target. Conversely, a dynamically admissible sequence can be statistically non-identifying. Sequence compatibility and identifiability are therefore logically distinct.

10.7 Sequence compatibility as a causal and predictive filter

The Temporal Sequence Compatibility Principle asks whether states used by an explanation can be embedded in at least one trajectory admitted by the governing equations and constraints. It is a necessary condition, not a complete causal theory. Reachability does not prove that a path was the actual cause, and the principle does not exclude common causes, proxies, reverse causation, or alternative mechanisms.

The generative extension makes the principle operational. A transition law is estimated or independently fixed on earlier states, frozen, and asked to predict an observation or population that did not participate in choosing the law. A localized sequence shock should appear as a structured residual. This procedure turns 'consistent with a history' into a stronger question: does the proposed history survive a genuinely prospective or locked holdout?

10.8 Comparative case analysis

The framework can now be compared across the paper's examples without assigning every case the same causal ordering. The table below lists the dominant role structure, not an exhaustive ontology.

Table 9. Comparative causal-role structure across the principal physical examples.

Case	Primary structural decomposition	Inherited state / constraint role	Main explanatory lesson
Small-angle pendulum	Geometry + gravity + restoring relation	Initial angle/velocity; fixed pivot and length	Constraint shapes the admissible path; inherited state selects the trajectory.

Case	Primary decomposition structural	Inherited state / constraint role	Main explanatory lesson
Isolated point charge	Point source + isotropic 3D spreading	Static case; ideal free-space domain	Source and geometry jointly generate the spherical inverse-square form.
Thin-wire current	3D field + carriers + slender conductor	Charge state; material boundary and circuit	Ambient field dimension and effective transport dimension need not coincide.
Cosmological NOS	Evolving geometry + cosmic source content + observational traces	Inherited cosmic state across epochs; selective transfer through ordinary dynamics, transfer functions, growth/damping, mode conversion, conservation, and observational accessibility; admissible sequence constraints.	A viable explanation must connect epoch-specific traces through one coherent pathway without treating cosmological history as a universal memory kernel.
Newtonian inertia	Flat inertial geometry + free continuation	Initial position and velocity; equal-time continuation must remain sequence-compatible with the inertial state	Geometry specifies the form of continuation; inherited state specifies the concrete motion; sampled positions form an arithmetic progression under uniform motion, and momentum provides a limiting case of minimal attenuation until new interaction occurs.
Relativistic free fall	Stress-energy → spacetime geometry → geodesic response	Initial event and four-velocity	Geometry mediates the response; inherited state selects the geodesic within the admissible family.
Galaxy dynamics	Baryonic source field + neutral residual response + independent channels	Flux constraint, selection, nuisance structure, and held-out temporal or population linkage	Predictive success does not by itself promote J_R/T_R to geometry or establish an ontology.

10.9 Why call the outcome a realized form?

The term realized form emphasizes that the explanandum is more specific than the general law but more structured than a single number. It includes symmetry, dimensionality, topology, spatial distribution, temporal organization, historical reachability, and scale-dependent regime. The framework therefore sits between law-level description and point prediction: it asks how a lawful possibility becomes this field pattern, this transport regime, this worldline, or this cosmological history.

The decomposition separates causal roles before recombining their physical effects; it does not claim that the world itself is separable into independent factors.

11. Philosophical Implications: Structure, Sequence, and Realization

11.1 Laws do not usually determine a unique concrete world-state

The first implication concerns the relation between laws and realized histories. A law such as the heat equation, Maxwell's equations, the geodesic equation, or the cosmological field equations does not generally select one unique realized configuration without additional state and constraint information.

Initial conditions and boundary or admissibility conditions are not optional afterthoughts if the explanatory target is a concrete phenomenon. In a historical process such as cosmology, a further question arises: whether the reconstructed states can be joined by one viable sequence. This observation does not weaken the law's universality; it clarifies what universality amounts to.

Philosophical discussions of determinism have long distinguished the uniqueness of evolution from the specification of initial data (Earman 1986). The present framework extends that point from temporal determinism to the broader constitution of physical form. A unique evolution from a given state may coexist with multiple possible states, domains, and boundaries.

11.2 Boundary conditions are explanatory, not merely computational

The second implication is that boundary and admissibility conditions can carry genuine explanatory content. In the heat equation, insulation or fixed temperature changes the allowed energy exchange; in conductor problems, the boundary helps explain the effective dimension of transport. In cosmology there may be no material wall, but a model still defines an admissible space of states and transitions through its geometry, field content, conservation relations, matching assumptions, and regime of validity. It would be misleading to treat these restrictions as merely mathematical conveniences when they determine which physical histories the model allows.

At the same time, a boundary condition can be effective rather than fundamental. A perfectly insulated wall is an idealization of suppressed microscopic exchange. The philosophical point is not that boundaries are ultimate entities but that the level of description determines which structural conditions are constitutive of the model.

11.3 Historical attenuation, retention, and temporal structural continuity

The past remains relevant through selectively retained or transformed state, not as an extra substance. Momentum can retain an earlier impulse, stress a manufacturing history, and cosmological perturbations information from earlier epochs. Dissipation erases some distinctions, conservation protects others, and coarse-graining changes which variables remain explanatory. An explicit memory kernel and a Markovian augmented state are alternative representations when they encode the same sufficient information. The philosophical object is therefore not the total past but the inherited structure that links a present state to a lawful history.

11.4 Physical laws and past-present-future sequence compatibility

Let Hadm be the histories satisfying the governing laws and constraints. A deterministic model may select one continuation from a complete state; a stochastic model assigns a distribution; a reversible law can constrain past and future without selecting a thermodynamic arrow. Sequence compatibility therefore means that states invoked together must belong to at least one admissible history. It is structural rather

than fatalistic. Causal explanation adds mechanism, asymmetry, intervention, or contrastive dependence, but it cannot require a transition forbidden by the dynamics.

11.5 A moderate, multi-level structural relationalism

The framework supports a moderate structural relationalism. Entities have scale-specific organization and causal capacities, yet their realized behavior also depends on geometrical, dynamical, historical, and boundary relations. Structure may be internal, as in molecular bonding or a lattice, or external, as in constraints and fields. This view preserves entities while rejecting explanations that isolate them from the relations through which their capacities are manifested; it does not require the stronger claim that objects are ontologically dispensable (Ladyman and Ross 2007; French 2014).

11.6 Symmetry, explanation, and source structure

Symmetry arguments are powerful because they eliminate arbitrary distinctions. An isolated point source in isotropic space cannot select one direction without additional structure. A homogeneous equation in a symmetric domain may admit symmetric solutions, although symmetry of the law alone does not guarantee symmetry of every solution: asymmetric initial or boundary conditions can break it. The explanatory role of symmetry therefore lies in the combined structure of law, source, state, and boundary (Brading and Castellani 2003).

This point corrects a common simplification. Spherical symmetry is not produced by spatial dimension alone. It arises from the relation between a centered source and an isotropic environment. Conversely, a homogeneous and isotropic space need not have a material center. Centered spherical symmetry and intrinsic spatial homogeneity are distinct structures.

11.7 Effective forms, limits, and scale

Straight lines, one-dimensional currents, and uniform temperature profiles may be exact within idealized models or effective within specified scales. A line can be a fundamental Euclidean object, a local tangent approximation, a zero-curvature limit, or the residual direction after transverse variations are suppressed. The philosophy of physical form must therefore be sensitive to asymptotic reasoning and scale-dependent representation (Batterman 2002; Butterfield 2011).

This does not reduce every macroscopic form to illusion. Effective forms can be robust and explanatorily indispensable even when they are not exact at microscopic scales. The wire's one-dimensional current is physically real as a macroscopic transport regime, though it supervenes on three-dimensional microscopic motion.

11.8 Decompression, recompression, and causal authority

Decomposition identifies which parts of an expression arise from convention, source, geometry, boundary, or inherited state. Controlled recompression then asks whether an effective coefficient is a new degree of freedom or a summary of known structure. In the galaxy case, J_R remains neutral because the baryonic geometry already enters g_{bar} ; only an independent $J_R \rightarrow J_I$ derivation can grant geometrical authority. The method does not predict an answer by itself. It locates where a new physical question can be asked without promoting a proxy too early.

11.9 Identifiability, prediction, geometry, and ontology

Analytical separability, empirical identifiability, predictive success, geometrical explanation, and ontological establishment form distinct evidential stages. Variables may be separated on paper yet remain observationally degenerate. Independent channels may identify a low-dimensional residual without showing that it is fundamental. A phenomenological response may predict held-out data while lacking an independent geometrical derivation. Even a successful geometrical model remains one empirically supported ontology among alternatives unless its distinctive consequences survive comparative tests.

Five-stage boundary: Analytically separable $\neq$ empirically identifiable $\neq$ predictively successful $\neq$ geometrically explained $\neq$ ontologically established.

The galaxy example makes the fourth distinction unavoidable. J_R and T_R are legitimate empirical response coordinates, but their notation does not authorize a geometrical reading. Promotion requires independent structure, not merely a good curve fit. The cosmological example adds an equally important warning: arranging cross-sectional samples by redshift does not create an individual history. A sequence claim requires a declared population transition and a selection-aware observation operator.

By keeping analytical separability, empirical identifiability, predictive success, geometrical explanation, and ontological establishment explicitly non-equivalent, the framework leaves a clear and auditable path for subsequent claim-specific, trigger-based multidimensional assessment without imposing a fixed set of evaluative dimensions in advance.

12. Scope and Limitations

12.1 Explanatory framework versus physical theory

The role-based framework and the ε_g reparameterization organize explanation; they are not substitutes for established field equations or for domain-specific causal models. Their contribution is strongest when compact notation combines quantities with different provenance and when separating those roles clarifies a model, exposes a hidden assumption, or defines an independently testable residual. If the standard theory already makes the relevant structure explicit and auditable, additional decomposition should remain minimal.

The symbols Γ , S , H , B , and I are therefore not assumed to be orthogonal variables. They are analytically separated explanatory coordinates or roles. Calling them orthogonal would require an explicit metric or inner product on the relevant model space; no such structure is assumed here. They can share physical degrees of freedom, influence one another, be linked by equality or dynamical constraints, or collapse into a smaller active set in a particular problem. Geometry is not a universal first cause, and inherited state is not a universal hidden variable. The adequacy of the decomposition is case-dependent.

12.2 Limits of the gravitational scalar representation

Because $\varepsilon_g = (4\pi G)^{-1}$, it is not an independently measured gravitational permittivity. A_r is likewise not a universal curvature scalar. Scalar recompression is admissible only when the observable-relevant response has been reduced without discarding active directional or nonlocal information. Kerr spacetime, for example, cannot be represented by changing a scalar area factor; a covariant metric, connection, or operator must remain explicit.

The identity is also dimension-specific: changing spatial dimension changes both sphere measure and the dimensions of the coupling. Finally, replacing the constant ε_g by $\varepsilon_g(x)$ in Einstein's equation is not a complete theory. The contracted Bianchi identity requires compatible dynamics and conservation, schematically $\nabla_\mu(T_{\mu\nu}/\varepsilon_g) = 0$. A variable response therefore needs additional fields or modified matter conservation, as in genuinely developed scalar–tensor theories (Brans and Dicke 1961; Carroll 2019).

12.3 Empirical status of the numerical cases

The SPARC computation is exploratory and was designed after the relevant galaxy relations were known; it is not a preregistered or fully blind confirmation. The tensor-lattice coefficients were fitted to the same Newtonian disk field whose compression error they measure. The flux-coupled finite-volume experiment, its convergence scan, the complementary conservation–sequence rank experiment, and the pseudo-longitudinal population audit were generated by declared toy laws. They test conservation, conditional generation, numerical convergence, model-form sensitivity, negative controls, rank, conditioning, and holdout design rather than the truth of those laws. A later empirical paper should freeze code, closures, sample cuts, quality controls, priors, derivative estimators, systematics, progenitor matching, and external validation data before evaluation.

12.4 Dark matter and cross-scale extrapolation

A successful J_R or T_R model in rotation curves would support a low-dimensional effective description. It would not exclude dark matter, transfer automatically to lensing or cosmology, or justify the same scalar or tensor in strong-field spacetime. Standard halo models, MOND-type theories, and residual-response models must be compared on the same samples with the same nuisance treatment and held-out

rules (Navarro, Frenk, and White 1997; Milgrom 1983). The manuscript therefore discusses a possible contribution to Newtonian-equivalent apparent mass, not the nonexistence of dark matter.

12.5 Three empirical meanings of $\varepsilon_{g0,s}$

A definition-derived ε_{g0} is merely the laboratory G_0 expressed as $1/(4\pi G_0)$. A model-conditioned cosmological estimate fits the same normalization within a declared GR/ Λ CDM or modified-gravity forward model and remains conditional on its microphysics and nuisance structure. A framework-independent measurement would have to isolate the response through observational channels whose independent rank contribution has been demonstrated. This paper reports neither a new constant nor such a measurement.

12.6 Limits of the sequence-compatibility interpretation

Sequence notation is an analytical representation, not a claim that nature is fundamentally discrete. A continuous differential equation may be sampled, discretized, or rewritten as a first-order system on an augmented state. Such Markovization changes the representation of the sufficient state; it does not erase the physical history by which that state was formed. The relevant invariant is whether the declared present state is sufficient for lawful continuation under the chosen model.

$$X_{t+1} = F(X_t; \theta) + \xi_t, \quad O_t = \mathcal{M}_{\text{obs}}(X_t) + \eta_t$$

Here $\mathcal{M}_{\text{obs}}$ denotes the observation operator; H remains reserved for the historical-inheritance role.

Nor does sequence compatibility imply universal determinism or a universal arrow of time. Probabilistic models define distributions over future states, and many microscopic equations are time-reversal symmetric even when coarse-grained boundaries support irreversible behavior. The framework uses past, present, and future to audit explanatory linkage inside a specified model; it does not offer an independent metaphysics of time.

Finally, a dynamically possible path need not be the actual cause of an observation. Causal authority still requires attention to source structure, interventions or counterfactuals where appropriate, alternative mechanisms, confounding, measurement provenance, and empirical discrimination. Sequence compatibility is a necessary filter on historical narratives, not a sufficient causal criterion.

13. Conclusion

Compact equations are indispensable for calculation, yet their efficiency often depends on compressing source, geometry, coupling, boundary, history, and observation into one expression. Explanatory decompression makes these roles auditable, after which the governing field equations, conservation laws, and admissibility conditions must be restored. Its main scientific gains are practical: it

exposes double counting, identifies null directions and missing measurements, distinguishes a fitted coefficient from its possible causes, and shows where an additional closure is required.

The SPARC stress test and Newtonian disk control show why that discipline matters. A neutral residual response can be frozen and tested on held-out galaxies, but the present families do not outperform the fixed radial-acceleration relation. Ordinary disk geometry already creates directional enhancement and spherical-estimator bias, so it cannot be counted again after entering the baryonic forward model. The flux-conserving generator supplies a stronger methodological example: a determinant-one response can redistribute fixed total flux and alter a finite-time configuration, while convergence failures still forbid an equilibrium or astronomical interpretation.

The synthetic conservation–sequence and pseudo-longitudinal experiments add a complementary lesson. Lawful sequence linkage can reduce admissible parameter freedom, but it does not itself provide empirical rank. Competing mass and response explanations may fit one equatorial rotation channel at one epoch; discrimination requires an independently informative forward channel—for example off-plane dynamics, lensing, independently calibrated source mass, or a genuinely held-out epoch. Promotion from J_R or T_R to geometry therefore requires independent structural derivation and cross-channel success, not rotation-curve fit alone.

The same workflow can be adapted to other complex physical phenomena, and cautiously to nonphysical dynamical systems, when domain-specific laws, transition operators, measurement models, and uncertainty structures are supplied. It is not a universal dynamical theory. It is a disciplined way to move from compact description to structured comparison, negative control, held-out prediction, and controlled recompression.

This is also a productive interaction between philosophy and science. Philosophy supplies distinctions among representation, prediction, explanation, and ontology; modelling forces those distinctions into forward equations, conservation residuals, convergence criteria, and possible failure. Scientific results then decide whether a conceptual separation has measurable consequences or remains non-identifying. Keeping analytical separability, empirical identifiability, predictive success, geometrical explanation, and ontological establishment non-equivalent leaves a clear trigger-based path for later multidimensional assessment without fixing its dimensions in advance.

Final methodological rule: Separate roles; restore field equations and conservation; identify likelihood rank before using priors; demand cross-channel and held-out sequence divergence; recompress only after the causal and ontological limits are explicit.

Appendix A. Reproducibility Structure of the Numerical Audits

A.1 Exploratory SPARC analysis

Public SPARC mass-model tables were filtered to SPARC quality classes $Q_{\text{SPARC}} = 1$ or 2. Thirty 80/20 splits were grouped by galaxy and scored by equal-galaxy log-velocity RMSE. Local-slope thresholds were exploratory. Confirmatory work must propagate distance, inclination, mass-to-light, gas, derivative, and covariance uncertainties on an independent sample.

A.2 Newtonian disk–tensor comparison

An infinitesimally thin exponential disk was integrated at 32–256 angular nodes. Spherical-shell flux closure, Legendre $l = 0 + 2$ and $l = 0 + 2 + 4$ compressions, integration tolerance, directional residuals, and spherical-estimator bias were audited. The fitted tensor compresses the same Newtonian field; it is not a new source of gravity.

A.3 Flux-conserving history-coupled generator

The axisymmetric finite-volume code evolves paired Newtonian and coupled branches from identical mass-ring states. It checks boundary flux, solver residual, particle step size, and diagnostic evolution while β relaxes toward a smoothed source-flattening target.

A.4 Convergence and robustness

Audits vary grid, time step, initial flattening, closure, particle count, random seed, domain, regularization length, and time horizon. Expanded domains and endpoint signs converge, whereas the original-domain and quasi-steady criteria fail; these failures remain part of the reported result.

A.5 Sequence and pseudo-longitudinal audits

Synthetic experiments report Jacobian rank, condition number, negative controls, held-out epochs, nonparallel-channel coverage, and selection-aware population matching. Recovery under the generating law validates the code and design, not the law in nature.

A.6 Confirmatory lock

A future release should freeze the J_R family and BTFR branch, sample cuts, nuisance model, derivative estimator, promotion gate, closure, boundary conditions, tolerances, progenitor matching, independent validation sample, code version, seeds, negative results, and AI-assistance disclosure before evaluation.

Acknowledgments and Declarations

Acknowledgments: None.

Funding: This research received no external funding.

Competing interests: The author declares no competing interests.

Data and code availability: The galaxy audit used the public SPARC database and its published mass-model tables. A self-contained computational archive contains the frozen inputs, executable scripts, expected outputs, exact dependency versions, SHA-256 manifest, and an independent verifier for the SPARC pressure test, Newtonian disk–tensor comparison, and flux-conserving generative model with its convergence audits. The assembled archive passed a fresh end-to-end rerun. Scripts and derived outputs have also been prepared for the complementary conservation–sequence rank experiment and pseudo-longitudinal population audit. The code, frozen numerical outputs, convergence audits, and verification materials supporting this study are available in the Geometry, History, and Constraint: Computational Reproducibility Archive, version 1.0.0, Zenodo. The version-specific DOI will be inserted upon publication; the all-versions DOI is <https://doi.org/10.5281/zenodo.22271471>

Use of generative AI: Generative AI assisted with organizing revision notes, drafting and debugging Python analysis code, numerical visualization, checking outputs, language editing, and initial document formatting. Numerical calculations were executed in reproducible Python environments. The author defined the research questions and assumptions, evaluated the physical and philosophical claims, reviewed the code and outputs, selected the reported results, and accepts full responsibility for the manuscript. AI is not listed as an author and is not treated as an independent source of evidence.

References

- Aristotle. (1984). *O_n the Heavens*. In J. Barnes (Ed.), *The Complete Works of Aristotle*. Princeton University Press.
- Batterman, R. W. (2002). *The Devil in the Details*. Oxford University Press.
- Brading, K., & Castellani, E. (Eds.). (2003). *Symmetries in Physics*. Cambridge University Press.
- Brans, C. H., & Dicke, R. H. (1961). Mach's principle and a relativistic theory of gravitation. *Physical Review*, 124, 925–935. <https://doi.org/10.1103/PhysRev.124.925>
- Butterfield, J. (2011). Less is different: Emergence and reduction reconciled. *Foundations of Physics*, 41, 1065–1135.
- Carroll, S. M. (2019). *Spacetime and Geometry*. Cambridge University Press.
- Carslaw, H. S., & Jaeger, J. C. (1959). *Conduction of Heat in Solids* (2nd ed.). Clarendon Press.
- Earman, J. (1986). *A Primer on Determinism*. D. Reidel.
- Einstein, A. (1915). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, 844–847.
- Einstein, A. (1916). Die Grundlage der allgemeinen Relativitätstheorie. *Annalen der Physik*, 49, 769–822.
- French, S. (2014). *The Structure of the World*. Oxford University Press.

- Ibata, R., et al. (2024). Charting the Galactic Acceleration Field. II. A global mass model of the Milky Way from the STREAMFINDER atlas of stellar streams detected in Gaia DR3. *Astrophysical Journal*, 967, 89. <https://doi.org/10.3847/1538-4357/ad382d>
- Izvekov, S. (2021). Mori–Zwanzig projection operator formalism. *Physical Review E*, 104, 024121.
- Ladyman, J., & Ross, D. (2007). *Every Thing Must Go*. Oxford University Press.
- Lee, M. H. (2000). Generalized Langevin equation and recurrence relations. *Physical Review E*, 62, 1769–1772.
- Lelli, F., McGaugh, S. S., & Schombert, J. M. (2016). SPARC: Mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves. *Astronomical Journal*, 152, 157. <https://doi.org/10.3847/0004-6256/152/6/157>
- Lelli, F., McGaugh, S. S., Schombert, J. M., Desmond, H., & Katz, H. (2019). The baryonic Tully–Fisher relation for different velocity definitions and implications for galaxy angular momentum. *Monthly Notices of the Royal Astronomical Society*, 484, 3267–3278. <https://doi.org/10.1093/mnras/stz205>
- McGaugh, S. S., Lelli, F., & Schombert, J. M. (2016). Radial acceleration relation in rotationally supported galaxies. *Physical Review Letters*, 117, 201101. <https://doi.org/10.1103/PhysRevLett.117.201101>
- Meyer, H., & Brandner, K. (2026). Weak-memory dynamics in discrete time. *Physical Review Letters*, 136, 220401. <https://doi.org/10.1103/kdhy-66b6>
- Milgrom, M. (1983). A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *Astrophysical Journal*, 270, 365–370.
- Navarro, J. F., Frenk, C. S., & White, S. D. M. (1997). A universal density profile from hierarchical clustering. *Astrophysical Journal*, 490, 493–508.
- Newton, I. (1999). *The Principia*. University of California Press. (Original work published 1687.)
- Pearl, J. (2009). *Causality* (2nd ed.). Cambridge University Press.
- Peebles, P. J. E. (1993). *Principles of Physical Cosmology*. Princeton University Press.
- Planck Collaboration. (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641, A6.
- Raue, A., et al. (2009). Structural and practical identifiability analysis by profile likelihood. *Bioinformatics*, 25, 1923–1929.
- Riess, A. G., et al. (2022). A comprehensive measurement of the local value of the Hubble constant. *Astrophysical Journal Letters*, 934, L7.
- Shin, T.-H., et al. (2018). The ellipticity of galaxy cluster halos from satellite galaxies and weak lensing. *Monthly Notices of the Royal Astronomical Society*, 475, 2421–2437. <https://doi.org/10.1093/mnras/stx3366>
- Vacchini, B. (2016). Generalized master equations leading to completely positive dynamics. *Physical Review Letters*, 117, 230401.
- Wald, R. M. (1984). *General Relativity*. University of Chicago Press.
- Weinberg, S. (2008). *Cosmology*. Oxford University Press.
- Woodward, J. (2003). *Making Things Happen*. Oxford University Press.